

Forcing the Π_n^1 -Uniformization Property

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Abstract

We generically construct a model in which the Π_3^1 -uniformization property is true, thus lowering the best known consistency strength from the existence of $M_1^\#$ to just ZFC. The forcing construction can be adapted to work over canonical inner models with Woodin cardinals, which yields, for the first time, universes where the Π_{2n}^1 -uniformization property holds, thus producing models which contradict the natural PD-induced pattern.

1 Introduction

The question of finding nicely definable choice functions for a definable family of sets is an old and well-studied subject in descriptive set theory; see, for instance, [13, 18]. The uniformization problem, first mentioned by N. Lusin in 1930 (see [15]), asks to find choice functions which lie at the same projective level as the set they aim to uniformize. Recall that for $A \subseteq 2^\omega \times 2^\omega$, a *uniformization* of A is a function

$$f : \text{pr}_1(A) \longrightarrow 2^\omega$$

such that

$$\text{graph}(f) \subseteq A.$$

Equivalently, for every $x \in \text{pr}_1(A)$ one has $(x, f(x)) \in A$.

Definition 1.1. *We say that a pointclass Γ has the uniformization property iff every element of Γ admits a uniformization in Γ .*

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It is a classical result due to M. Kondo [14] that lightface Π_1^1 -sets do have the uniformization property, this also yields the uniformization property for Σ_2^1 -sets. This is as much as ZFC can prove about uniformization. In the constructible universe L , as shown by J. Addison in [1], Σ_n^1 does have the uniformization property for $n \geq 3$, which follows from the existence of a Σ_2^1 -good wellorder of the reals; for background on projective wellorders see also [2, 4]. Thus the Π_n^1 -uniformization fails for $n \geq 3$. On the other hand, by the celebrated results of Y. Moschovakis (see [17], Theorem 1), Δ_{2n}^1 -projective determinacy implies Π_{2n+1}^1 -uniformization, yet the determinacy assumption exceeds in logical strength ZFC. It is known due to H. W. Woodin (see [19]), that Δ_2^1 -projective determinacy implies the existence of $M_1^\#$, hence yields an inner model with a Woodin cardinal. As with other regularity properties of the reals like Lebesgue measurability or Baire property, which both follow from PD as well, it is natural to ask whether the Π_3^1 -uniformization property bears large cardinal strength as well. We shall answer it negatively.

Theorem. *There is a generic extension of L in which the Π_3^1 -uniformization property is true.*

The proof can be adapted such that it applies to canonical inner models with Woodin cardinals. This can be used to obtain better lower bounds in terms of consistency strength for the Π_n^1 uniformization property for odd n . For even n we can produce for the first time models where the Π_n^1 uniformization property holds true.

Theorem. *Let M_n be the canonical inner model with n Woodin cardinals. Then there is a generic extension of M_n in which the Π_{n+3}^1 uniformization property holds true.*

Since the first version of this paper was written, the method developed here has led to several further constructions. A variant combines Π -uniformization with a projective well-order of the reals under CH [7]; another combines global Σ -uniformization with a large continuum and a projective well-order [8]; and the local Delfino constructions combine related coding and allowability methods with projective regularity, definable well-orders, and adjacent Σ - or Π -uniformization [10, 9]. These later papers should be viewed as refinements and applications of the forcing machinery introduced here. The present paper isolates the original fixed-point and allowability mechanism needed to force projective Π -uniformization without assuming the corresponding determinacy strength.

Questions concerning the forceability of (local) consequences of PD do have a long tradition in set theory. There is a vast body of literature concerning the forcability of local levels of the projective hierarchy satisfying (Boolean combinations of) the Baire property, the perfect set property or Lebesgue measurability; see for instance [5]. There has been very little progress in the

past, however, concerning similar questions for the separation, the reduction and the uniformization property. Indeed, even the question of whether one can force the Σ_3^1 -separation property, which is the weakest of said properties, remained an open problem for 50 years (see [16], Problem 3029).

This article continues this line of research and introduces the forcing template on which the subsequent work mentioned above builds. It is organized as follows: in the preliminaries section, we briefly introduce the forcings which we will use in the proof and produce a generic extension W of L which will be a well-suited ground model for our needs. We then start to prove the theorems from above. The main idea is to turn the problem of finding a partial order $\mathbb{P}$ which forces the Π_3^1 -property into a fixed point problem. We shall define a derivation operator which acts on a specific set of forcing iterations of length ω_1 . This operator will be applied transfinitely often, and will produce better and better approximations to the set of forcings we actually want to use in the end. The process is shown to converge in that eventually a fixed point $\mathcal{P}$, i.e. a suitable set of forcings is reached. Forcings which belong to this fixed point $\mathcal{P}$ allow for a certain, seemingly self-referential definition of an ω_1 -length iteration which forces the Π_3^1 -uniformization property over W . We then follow up, to alter the said process such that it becomes applicable to the canonical inner models M_n with n Woodin cardinals.

There are some similarities to [6], in particular the two proofs rely on a similar ground model W , which is a generic extension of L , and use a similar coding method which relies on a suitably chosen ω_1 -sequence of ω_1 -Suslin trees. However a more straightforward application of the ideas of [6] will fail badly to produce a model of the Π_3^1 -uniformization property. As a consequence, a solution has to necessarily introduce several new ideas in order to succeed. The presentation of those is the goal of this paper.

2 Preliminaries

2.1 Notation

The notation we use will be mostly standard, we hope. We write $\mathbb{P} = (\mathbb{P}_\alpha : \alpha < \gamma)$ for a forcing iteration of length γ with initial segments $\mathbb{P}_\alpha$. The α -th factor of the iteration will be denoted with $\mathbb{P}(\alpha)$. Note here that we drop the dot on $\mathbb{P}(\alpha)$, even though $\mathbb{P}(\alpha)$ is in fact a $\mathbb{P}_\alpha$ -name of a partial order.

2.2 The forcings which are used

The forcing notions used in the construction are standard. The preparatory extension is obtained solely by adding branches through a fixed sequence of Suslin trees, and the later coding uses the ground-model version of the ω_1 -Cohen forcing together with Jensen–Solovay almost disjoint coding [12].

Recall that a Suslin tree is a normal tree of height ω_1 with countable levels and without an uncountable chain or antichain. Whenever S is a Suslin tree, we also write S for the forcing whose conditions are the nodes of S , ordered by end-extension. Thus forcing with S adds a cofinal branch through S .

Definition 2.1. Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be a sequence of Suslin trees. We call $\vec{S}$ independent if, for every nonempty finite set $e \subseteq \kappa$, the product

$$\prod_{\xi \in e} S_\xi$$

is again a Suslin tree.

For $A \subseteq \kappa$ let

$$\text{Br}_A(\vec{S}) = \prod_{\xi \in A}^{\text{fs}} S_\xi$$

denote the finite-support product of the corresponding branch forcings. When the sequence is fixed, we simply write Br_A .

Lemma 2.2. Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be an independent sequence of Suslin trees, where $\kappa \leq \omega_1$.

1. For every $A \subseteq \kappa$, the forcing Br_A is c.c.c.
2. If $A \subseteq B \subseteq \kappa$, then Br_A is a complete subforcing of Br_B , and

$$\text{Br}_B \cong \text{Br}_A \times \text{Br}_{B \setminus A}.$$

3. If $\xi \notin A$, then

$$\text{Br}_A \Vdash "S_\xi \text{ is a Suslin tree}."$$

Proof. For (1), let $\{p_\alpha \mid \alpha < \omega_1\}$ be a family of conditions in Br_A . After thinning it out, the supports form a Δ -system with finite root r , and the restrictions $p_\alpha \restriction r$ belong to the finite product $\prod_{\xi \in r} S_\xi$. By independence this finite product is Suslin, hence c.c.c., so two of the root restrictions are compatible. Since the remaining parts of their supports are disjoint, the corresponding two conditions in Br_A are compatible.

The factorization in (2) is the usual coordinatewise factorization of a finite-support product. For (3), observe that

$$\text{Br}_A \times S_\xi \cong \text{Br}_{A \cup \{\xi\}},$$

which is c.c.c. by (1). The standard product characterization of preservation of a Suslin tree now yields the assertion. $\square$

We next recall the almost disjoint coding forcing due to Jensen and Solovay [12]. We identify subsets of ω with their characteristic functions. Let $D = \langle d_\alpha \mid \alpha < \omega_1 \rangle$ be an almost disjoint family of infinite subsets of ω , and let $X \subseteq \omega_1$.

Definition 2.3. *The almost disjoint coding forcing $\mathbb{A}_D(X)$ consists of pairs $(r, R) \in [\omega]^{<\omega} \times [D]^{<\omega}$. We put $(s, S) \leq (r, R)$ if and only if*

1. $r \subseteq s$ and $R \subseteq S$;
2. whenever $\alpha \in X$ and $d_\alpha \in R$, then $s \cap d_\alpha = r \cap d_\alpha$.

If $G \subseteq \mathbb{A}_D(X)$ is generic and $x_G = \bigcup \{r : (r, R) \in G\}$, then

$$\alpha \in X \iff x_G \cap d_\alpha \text{ is finite.}$$

Throughout the paper, D denotes the canonical L -definable almost disjoint family obtained by recursively choosing the $<_L$ -least infinite subset of ω almost disjoint from all previously chosen members.

Lemma 2.4. *For every $X \subseteq \omega_1$, the forcing $\mathbb{A}_D(X)$ is Knaster. In particular it is c.c.c. and preserves every ground-model Suslin tree.*

Proof. The Knaster property follows from the usual Δ -system argument applied to the finite side conditions. If T is a Suslin tree, then the product of a Knaster forcing with the c.c.c. forcing T is c.c.c. Hence $\mathbb{A}_D(X) \times T$ is c.c.c., which is equivalent to preservation of T as a Suslin tree. $\square$

Finally, we write

$$\mathbb{C}_L(\omega_1) = ({}^{<\omega_1}2)^L$$

for the L -computed ω_1 -Cohen forcing, ordered by end-extension. Thus conditions are countable binary sequences belonging to L . Whenever this forcing is used in an outer model of L , the notation refers to this fixed L -poset rather than to a recomputation in the ambient universe.

2.3 The ground model W of the iteration

We now define the ground model over which the main iteration will be carried out. We start with the canonical independent Suslin sequence already present in L and add a cofinal generic branch through every member of that sequence.

The following lemma gives the simultaneous form of Jensen's diamond construction which we use throughout the paper; see also [11]. We use the levelwise product of finitely many trees. This is a dense suborder of the corresponding finite forcing product.

Lemma 2.5. *There is a canonical L -definable sequence*

$$\vec{S} = \langle S_\xi \mid \xi < \omega_1 \rangle$$

of independent normal Suslin trees. The construction is given by a recursion which is uniform over L_{ω_1} .

Proof. Work in L . Fix the canonical L -diamond sequence $\langle a_\alpha \mid \alpha < \omega_1 \rangle$ and the canonical L -coding of finite sets of ordinals, countable trees, and subsets of their finite products. We recursively construct coherent countable approximations $\vec{S}^\alpha = \langle S_\xi^\alpha \mid \xi < \alpha \rangle$ such that each S_ξ^α is a normal splitting tree of height α and $S_\xi^\beta = S_\xi^\alpha \upharpoonright \beta$ whenever $\xi < \beta < \alpha$. At limit stages we take unions.

It remains to describe the passage from α to $\alpha + 1$. For old coordinates $\xi < \alpha$, successor stages are handled by putting countably many immediate successors above each node on the top level. Suppose now that α is a nonzero limit. If a_α does not decode as a pair (e, A) , where $e \in [\alpha]^{<\omega} \setminus \{\emptyset\}$ and A is a maximal antichain in the product

$$P_e^\alpha = \bigcup_{\delta < \alpha} \prod_{\xi \in e} (S_\xi^\alpha)_\delta,$$

then we use the $<_L$ -least countable family of cofinal branches through each old tree which covers all old nodes, and put one new top node above each of these branches.

If a_α does decode such a pair (e, A) , we choose the branch families for the coordinates in e simultaneously so as to seal A . Concretely, by a countable genericity argument in $\prod_{(\xi, n) \in e \times \omega}^{\text{fs}} S_\xi^\alpha$, choose cofinal branches $b_{\xi, n}$, $\xi \in e$, $n < \omega$, such that they cover all old nodes and such that, for every $\nu : e \rightarrow \omega$, the tuple of new top nodes placed above $\langle b_{\xi, \nu(\xi)} \mid \xi \in e \rangle$ extends some member of A . The required dense sets are just the usual ones ensuring cofinality, coverage of old nodes, and the displayed sealing requirement; density of the last kind uses maximality of A in P_e^α . For $\xi \notin e$ we again use the default $<_L$ -least covering family. Finally, the new coordinate α is initialized as the canonical finite-support binary tree of height $\alpha + 1$. This completes the recursion. Put $S_\xi = \bigcup_{\alpha < \omega_1} S_\xi^\alpha$. Then each S_ξ is a normal splitting tree of height ω_1 with countable levels.

We show that the sequence is independent. Fix a finite nonempty $e \subseteq \omega_1$ and let

$$P_e = \bigcup_{\alpha < \omega_1} \prod_{\xi \in e} (S_\xi)_\alpha$$

be the product. This is dense in the usual finite product of the corresponding tree forcings. Let $A \subseteq P_e$ be a maximal antichain in L . There is a club of $\alpha > \sup(e)$ such that $A \cap (P_e \upharpoonright \alpha)$ is maximal in $P_e \upharpoonright \alpha$, and, by coherence of the coding, a club on which the fixed code $x_{e, A}$ for (e, A) restricts to a code

for $(e, A \cap (P_e \restriction \alpha))$. By $\diamond$, choose a limit α in the intersection of these clubs with $a_\alpha = x_{e,A} \cap \alpha$. At stage α the construction seals $A \cap (P_e \restriction \alpha)$. Hence every node on level α , and therefore every later node, extends a member of this initial antichain. Since A itself is an antichain, $A = A \cap (P_e \restriction \alpha)$, so A is countable. Thus P_e has no uncountable antichain. A cofinal branch through P_e would, by normal splitting, yield an uncountable antichain by choosing at successor levels nodes incompatible with the branch. Hence P_e is Suslin. Since e was arbitrary, $\vec{S}$ is independent. The construction is canonical from $<_L$, the diamond sequence, and the fixed coding, and is therefore uniform over L_{ω_1} . $\square$

Lemma 2.6. *Let M be a transitive uncountable model of*

$$\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$$

such that

$$\omega_1^M = \omega_1^L = \omega_1.$$

There is a Σ_1 -formula

$$\vartheta_{\text{Sus}}(t, \gamma, \eta, \omega_1),$$

independent of M , which defines over M the relation

$$\{(t, \gamma, \eta) : \gamma, \eta < \omega_1 \text{ and } t \in (S_\eta)_\gamma\}.$$

Equivalently, there is a formula $\text{CanSuslin}_L(X, \omega_1)$, which may be taken to be $\Sigma_1(\omega_1)$ over $H(\omega_2)$, such that in every ω_1 -preserving outer model U of L ,

$$U \models \text{CanSuslin}_L(X, \omega_1) \iff X = \vec{S}.$$

In particular, the canonical sequence $\vec{S}$ is computed in the same way in all the outer models considered below.

Proof. By Lemma 2.5, the sequence $\vec{S}$ is obtained by a canonical recursion over L_{ω_1} , and the corresponding level relation is $\Sigma_1(\omega_1)$ -definable over L_{ω_1} . If M is transitive, satisfies $\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$, and has $\omega_1^M = \omega_1^L$, then M computes L_{ω_1} correctly. Hence M computes the same canonical diamond sequence, the same coding apparatus, and therefore the same recursive construction of $\vec{S}$. The same Σ_1 -formula $\vartheta_{\text{Sus}}(t, \gamma, \eta, \omega_1)$ consequently defines over M the relation $t \in (S_\eta)_\gamma$.

The formulation by $\text{CanSuslin}_L(X, \omega_1)$ is obtained by saying that X is the union of the unique system produced by this canonical recursion. Since every ω_1 -preserving outer model of L has the same L_{ω_1} , it satisfies this formula exactly for the true sequence $\vec{S}$. $\square$

Let

$$\text{Br} = \prod_{\xi < \omega_1}^{\text{fs}} S_\xi$$

be the finite-support branch product, computed in L , and let $H \subseteq \text{Br}$ be L -generic. For every $\xi < \omega_1$ set

$$b_\xi = \bigcup \{p(\xi) : p \in H \text{ and } \xi \in \text{dom}(p)\}.$$

Then b_ξ is a cofinal L -generic branch through S_ξ . In this paper, to *destroy* a Suslin tree always means to add such a cofinal generic branch through it. We define

$$W = L[H].$$

Theorem 2.7. *The model W has the following properties.*

1. *Br is a c.c.c. forcing of size $\aleph_1$. Hence W preserves all cardinals and cofinalities of L , and $W \models \text{CH}$.*
2. *Every S_ξ is destroyed in W , witnessed by the branch b_ξ .*
3. *For $A \subseteq \omega_1$ in L , let $H_A = H \cap \text{Br}_A$. Then H_A is Br_A -generic over L , and*

$$L[H_A] \models \text{“}S_\xi \text{ has a cofinal branch”} \iff \xi \in A.$$

4. *Every real in W belongs to $L[H_A]$ for some countable $A \subseteq \omega_1$ belonging to L .*

Proof. The first assertion follows from Lemma 2.2. Since $L \models \text{CH}$ and Br has size $\aleph_1$, there are only $\aleph_1$ many nice Br -names for reals, and therefore W still satisfies CH . The definition of H and the dense sets requiring a condition to reach a prescribed level of a prescribed coordinate give (2).

For (3), use the factorization

$$\text{Br} \cong \text{Br}_A \times \text{Br}_{\omega_1 \setminus A}.$$

If $\xi \in A$, then $b_\xi \in L[H_A]$. If $\xi \notin A$, then Lemma 2.2 shows that Br_A preserves S_ξ as a Suslin tree, so no cofinal branch through S_ξ belongs to $L[H_A]$.

Finally, let $\dot{x}$ be a nice Br -name for a real. The antichains occurring in $\dot{x}$ are countable, and the union of the finite supports of all conditions appearing in them is therefore a countable set $A \subseteq \omega_1$ in L . Thus $\dot{x}$ is already a Br_A -name, proving (4). $\square$

The relevant inner models of W used below are obtained by retaining selected branch coordinates from H .

Coding forcing Let $x \in W$ be a real and let $\eta < \omega_1$. Write $\mathbb{C}_L(\omega_1)_\eta$ for a tagged copy of $\mathbb{C}_L(\omega_1)$, and let $\dot{g}_\eta$ be its canonical name for the generic subset of ω_1 . We now describe a canonical $\mathbb{C}_L(\omega_1)_\eta$ -name $\dot{Y}(\check{x}, \dot{g}_\eta)$ for a subset of ω_1 . The coding forcing is the two-step iteration

$$\text{Code}(x, \eta) := \mathbb{C}_L(\omega_1)_\eta * \dot{\mathbb{A}}_D(\dot{Y}(\check{x}, \dot{g}_\eta)).$$

For the definition of Y , let $g \subseteq \omega_1$ be $\mathbb{C}_L(\omega_1)_\eta$ -generic over W , and identify g with its characteristic function in ${}^{\omega_1}2$. Let

$$\varrho_L : ({}^{<\omega_1}2)^L \longrightarrow \omega_1$$

be the canonical L -definable bijection. We use ϱ_L and g to form the set $h \subseteq \omega_1$ of indices of the ω -blocks of $\vec{S}$ on which the characteristic function of x will be written. Put

$$h := \{\varrho_L(g \upharpoonright \alpha) : \alpha < \omega_1\}$$

and

$$\begin{aligned} A := & \{\omega\gamma + 2n : \gamma \in h, n \notin x\} \\ & \cup \{\omega\gamma + 2n + 1 : \gamma \in h, n \in x\}. \end{aligned}$$

Let $X \subseteq \omega_1$ be chosen canonically so that it codes the following objects:

- the set $A \subseteq \omega_1$;
- the family $\langle b_\beta \mid \beta \in A \rangle$, where b_β is the branch through S_β added by the β -th coordinate of H . In particular, b_β is L -generic for the forcing S_β .

If $L_\theta[X]$ is some uncountable ZFC^- -model, and if we assume that $\gamma \in h$ then $L_\theta[X]$ can read off x via looking at the ω -block of $\vec{S}$ -trees starting at γ and determine which tree has an ω_1 -branch in $L_\theta[X]$. Thus we say that x is coded into $\vec{S}$ at the ω -block starting at γ in such a situation. To be more specific $L_\theta[X]$ satisfies the following formula:

(*) $_{\gamma, x}$ $n \in x$ if and only if $S_{\omega \cdot \gamma + 2n+1}$ has an ω_1 -branch, and $n \notin x$ if and only if $S_{\omega \cdot \gamma + 2n}$ has an ω_1 -branch.

Indeed if $n \notin x$ then we added a cofinal branch through $S_{\omega \cdot \gamma + 2n}$. If on the other hand $S_{\omega \cdot \gamma + 2n}$ does not have an ω_1 -branch in $L_\theta[X]$ then we must have added an ω_1 -branch through $S_{\omega \cdot \gamma + 2n+1}$ as we always add an ω_1 -branch through either $S_{\omega \cdot \gamma + 2n+1}$ or $S_{\omega \cdot \gamma + 2n}$ and adding branches through some S_α 's will not affect that some S_β remain Suslin in $L[X]$, as $\vec{S}$ is independent.

We now apply a localization argument inspired by David's trick¹ to this situation. We reorganize the information coded by $X \subset \omega_1$ as a set $Y \subset \omega_1$ as follows. It is clear that any transitive, $\aleph_1$ -sized ZFC^- model N of the form $N = L_\theta[X]$ will be able to first define $\vec{S}$ correctly and also correctly decode out of X all the information regarding x being coded at each ω -block of $\vec{S}$ starting at every $\gamma \in h$. Consequently, if we code the model $(N, \in)$ which contains X , as a set $X_N \subset \omega_1$, then for any uncountable β such that $L_\beta[X_N] \models \text{ZFC}^-$:

$L_\beta[X_N] \models$ "The model decoded out of X_N satisfies $(*)_{\gamma,x}$ for every $\gamma \in h$ ".

In particular there will be an $\aleph_1$ -sized ordinal β as above and we can fix a club $C \subset \omega_1$ and a sequence $(M_\alpha : \alpha \in C)$ of countable elementary submodels of $L_\beta[X_N]$ such that

$$\forall \alpha \in C (M_\alpha \prec L_\beta[X_N] \wedge M_\alpha \cap \omega_1 = \alpha)$$

Now let the set $Y \subset \omega_1$ code the pair (C, X_N) such that the odd entries of Y should code X_N and if $E(Y)$ denotes the set of even entries of Y and $\{c_\alpha : \alpha < \omega_1\}$ is the enumeration of C then

1. $E(Y) \cap \omega$ codes a well-ordering of type c_0 .
2. $E(Y) \cap [\omega, c_0) = \emptyset$.
3. For all β , $E(Y) \cap [c_\beta, c_\beta + \omega)$ codes a well-ordering of type $c_{\beta+1}$.
4. For all β , $E(Y) \cap [c_\beta + \omega, c_{\beta+1}) = \emptyset$.

The following formula $\sigma(v_0)$ is now true for our real x :

$\sigma(x) \equiv$ For any countable transitive model M of " ZFC^- and $\aleph_1$ exists" such that $\omega_1^M = (\omega_1^L)^M$ and $Y \cap \omega_1^M \in M$, M can construct its version of the universe $L[Y \cap \omega_1^M]$, and the latter will see that there is an $\aleph_1^M$ -sized transitive model $N \in L[Y \cap \omega_1^M]$ which models $(*)_{\gamma,x}$ for $\aleph_1^M$ -many $\gamma \in M$.

Thus we have a local version of the property $(*)_{\gamma,x}$.

We have finally defined the desired set Y . The second step of $\text{Code}(x, \eta)$ is

$$\mathbb{A}_D(Y),$$

the Jensen–Solovay almost disjoint coding forcing from Definition 2.3, computed with the fixed canonical family $D \in L$. It codes the set Y into one real r . This forcing only depends on the subset of ω_1 we code, thus $\mathbb{A}_D(Y)$ will

¹See [3] for the original argument, where the strings in Jensen's coding machinery are altered so that certain unwanted universes are destroyed. This destruction is emulated in our context below.

be independent of the surrounding universe in which we define it, as long as it has the right ω_1 and contains the set Y .

The effect of the coding forcing $\text{Code}(x, \eta)$ is that it generically adds a real r such that

$\Psi(r, x) \equiv$ For any countable, transitive model M of “ ZFC^- and $\aleph_1$ exists”, such that $\omega_1^M = (\omega_1^L)^M$ and $r \in M$, M can construct its version of $L[r]$, denoted by $L[r]^M$, which in turn thinks that there is a transitive ZFC^- -model N of size $\aleph_1^M$ such that N believes $(*)_{\gamma, x}$ for an $\aleph_1^M$ -sized set of ordinals γ .

Indeed, if r and M are as above, then M and $L[r]^M$ compute the canonical almost disjoint family D correctly below ω_1^M . As a consequence, $L[r]^M$ will contain the set $Y \cap \omega_1^M$, where $Y \subset \omega_1$ is as in $\sigma(x)$. So in $L[Y \cap \omega_1^M]$, there is an $\aleph_1^M$ -sized, transitive N which models $(*)_{\gamma, x}$ for $\aleph_1^M$ -many $\gamma \in M$, as claimed.

Note that $\Psi(r, x)$ is a Π_2^1 -formula in the parameters r and x , as the set $h \cap M \subset \omega_1^M$ is coded into r . We say in the above situation that r witnesses that the real x *is written into $\vec{S}$* , or that x *is coded into $\vec{S}$* .

To summarize, for every real x and every $\eta < \omega_1$, the forcing $\text{Code}(x, \eta)$ adds a real r such that the Π_2^1 -formula $\Psi(r, x)$ holds in the resulting extension of W .

We let $\Phi(x)$ denote the Σ_3^1 -statement “there is a real witnessing that x is coded into $\vec{S}$ ” that is

$$\Phi(x) := \exists r \Psi(r, x).$$

Via spelling out $\Psi(r, x)$ we see that $\Phi(x)$ is the following formula:

$$\begin{aligned} \Phi(x) \equiv \exists r \forall M \big(& M \text{ is countable and transitive, } M \models \text{ZFC}^-, \\ & \omega_1^M = (\omega_1^L)^M \rightarrow M \models \text{“} L[r] \text{ sees that } r \text{ codes} \\ & \text{a transitive } \aleph_1^M\text{-sized } \text{ZFC}^-\text{-model } N' \\ & \text{such that } N' \text{ satisfies } (*)_{\gamma, x} \text{ for } \aleph_1^M\text{-many } \gamma \text{”} \big). \end{aligned}$$

The projective and local statement $\Phi(x)$, if true, will determine how certain inner models of the surrounding universe will look like with respect to branches through $\vec{S}$. That is to say, if we assume that $\Psi(r, x)$ holds then r also witnesses that the assertion of $\Psi(r, x)$ remains true even if we replace the countable, transitive models M in the statement of $\Psi(r, x)$ with any transitive model M of the theory “ $\text{ZFC}^- + \aleph_1$ exists and $\aleph_1 = \aleph_1^L$ ”. In other words we can drop the assumption on the countability of M in $\Psi(r, x)$, provided $\Psi(r, x)$ is true. Indeed if we assume that there would be an uncountable, transitive M which models ZFC^- and “ $\aleph_1$ exists” and $\omega_1 = \omega_1^L$, which additionally contains r and which witnesses that the conclusion of $\Psi(r, x)$ is false. Then by Löwenheim-Skolem, there would be a countable $N \prec M$, $r \in N$ which we can transitively collapse to obtain the transitive $\bar{N}$. But $\bar{N}$ would witness

that the conclusion of $\Psi(r, x)$ is not true for every countable, transitive model, which is a contradiction to our assumption that $\Psi(r, x)$ is true.

Consequently, the real r carries enough information that the universe $L[r]$ will see that certain trees from $\vec{S}$ have branches in that $L[r]$ will decode, with the help of the almost disjoint family $D \in L$ a set $h \subset \omega_1$ such that

$$n \in x \Rightarrow L[r] \models \forall \gamma \in h \text{ (``} S_{\omega_\gamma+2n+1} \text{ has an } \omega_1\text{-branch'')}.$$

and

$$n \notin x \Rightarrow L[r] \models \forall \gamma \in h \text{ (``} S_{\omega_\gamma+2n} \text{ has an } \omega_1\text{-branch'')}.$$

Indeed, the universe $L[r]$ will see that there is a transitive model N of “ $\text{ZFC}^- + \aleph_1$ exists and $\aleph_1 = \aleph_1^L$ ” which believes $(*)_{\gamma, x}$ for every $\gamma \in h \subset \omega_1$, the latter being coded into r . But by upwards Σ_1 -absoluteness, and the fact that N can compute $\vec{S}$ correctly, if N thinks that some tree in $\vec{S}$ has a branch, then $L[r]$ must think so as well.

The definition of the forcing $\text{Code}(x, \eta)$ exhibits a significant degree of absoluteness. Specifically, its definition is independent of the universe in which it is computed, provided that the universe has the correct ω_1 and contains the real x . We will leverage this property shortly.

We do not iterate the full forcings $\text{Code}(x, \eta)$. Instead, we separate their two components. We first add one global ω_1 -Cohen reservoir and then, over the resulting extension, form a finite-support iteration of the almost disjoint coding forcings.

2.4 Allowable forcings

The forcing $\text{Code}(u, \eta)$ described above has two components. Its ω_1 -Cohen part chooses a coding area, while its Jensen–Solovay part writes the resulting set $Y \subseteq \omega_1$ into a real. In the applications below, u will be the fixed recursive real code of a triple (x, y, m) . We write

$$\dot{Y}(\dot{x}, \dot{y}, \dot{m}, \dot{g})$$

for the canonical name of the set Y obtained by applying the preceding coding construction to the recursive code of $(\dot{x}, \dot{y}, \dot{m})$ and the Cohen generic $\dot{g}$.

In an allowable forcing the two components of the coding forcing are organized separately. Put

$$\mathbb{C}^{\text{area}} := \prod_{\eta < \omega_1}^{\text{cs}} \mathbb{C}_L(\omega_1).$$

After passing to the standard dense partial-function presentations and fixing a bijection between ω_1 and $\omega_1 \times \omega_1$, the forcing $\mathbb{C}^{\text{area}}$ is isomorphic to one

copy of $\mathbb{C}_L(\omega_1)$. We retain the product presentation because its coordinate generics

$$\langle \dot{g}_\eta \mid \eta < \omega_1 \rangle$$

supply the coding areas used by the later iteration. For every $\eta < \omega_1$, the generic $\dot{g}_\eta$ determines the area

$$\dot{h}_\eta := \{\varrho_L(\dot{g}_\eta \restriction \xi) : \xi < \omega_1\}.$$

As computed in L , the forcing $\mathbb{C}^{\text{area}}$ is σ -closed. Since $W = L[H]$ is obtained from L by the c.c.c. forcing Br , Easton's lemma implies that

$$W \models \text{"}\mathbb{C}^{\text{area}} \text{ is } \omega\text{-distributive."}$$

See [11, Chapter 15]. Consequently, $\mathbb{C}^{\text{area}}$ adds no reals over W . The same remains true after permuting its coordinates. Thus the product presentation provides ω_1 many coding areas without adding reals at the reservoir step.

Over the reservoir extension we form a finite-support iteration of almost disjoint coding forcings. Schematically, at a stage β , the bookkeeping supplies names

$$\dot{x}_\beta, \quad \dot{y}_\beta, \quad \dot{m}_\beta.$$

Independently of the generic filter, we fix in L an injective assignment

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle$$

of reservoir coordinates to the stages of the iteration. For every $\beta \leq \delta$, put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

The bookkeeping data used at stage β will be represented by names whose reservoir support is contained in U_β . Since $\vec{\eta} \in L$, the same assignment is available in every branch inner model of W . The restriction on the names does not remove any possible real value: after the c.c.c. forcing which produces W and the c.c.c. almost disjoint tail below β , the unused part of the L -computed σ -closed reservoir is ω -distributive by Easton's lemma. The next factor is

$$\dot{\mathbb{A}}_\beta := \mathbb{A}_D(\dot{Y}(\dot{x}_\beta, \dot{y}_\beta, \dot{m}_\beta, \dot{g}_{\eta_\beta})).$$

If

$$\dot{\mathbb{R}}_\delta = \bigast_{\beta < \delta}^{\text{fs}} \dot{\mathbb{A}}_\beta,$$

then the full forcing is

$$\mathbb{P}_\delta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\delta.$$

Thus the Cohen reservoir is added only once. After this first step, only the almost disjoint coding forcings are iterated, with finite support.

Let $K * G$ be $\mathbb{P}_\delta$ -generic over W . For $\beta < \delta$, put

$$u_\beta := \langle \dot{x}_\beta^{K*G}, \dot{y}_\beta^{K*G}, \dot{m}_\beta^{K*G} \rangle.$$

We call $\{u_\beta : \beta < \delta\}$ the set of reals intentionally coded by the iteration. It is essential that this iterative process does not inadvertently code reals which were not explicitly intended. Lemma 2.12 will show that, in the final extension,

$$\Phi(u) \iff u = u_\beta \text{ for some } \beta < \delta.$$

Definition 2.8. Let $\delta < \omega_1$, let

$$F = \langle (\dot{x}_\beta, \dot{y}_\beta, \dot{m}_\beta) \mid \beta < \delta \rangle \in W$$

be a bookkeeping function, and let

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle \in L$$

be an injective sequence of ordinals below ω_1 . For $\beta \leq \delta$, put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

An allowable iteration over the reservoir of length δ relative to $(F, \vec{\eta})$ is a sequence

$$\mathbb{P} = \langle \mathbb{P}_\beta \mid \beta \leq \delta \rangle, \quad \mathbb{P}_\beta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\beta,$$

defined recursively as follows.

1. $\dot{\mathbb{R}}_0$ is the trivial forcing. Thus

$$\mathbb{P}_0 = \mathbb{C}^{\text{area}}.$$

2. Suppose that $\dot{\mathbb{R}}_\beta$ has been defined. By induction, $\dot{\mathbb{R}}_\beta$ is the canonical lift of a $\mathbb{C}_{U_\beta}^{\text{area}}$ -name. Put

$$\mathbb{P}_\beta^{U_\beta} := \mathbb{C}_{U_\beta}^{\text{area}} * \dot{\mathbb{R}}_\beta.$$

Then $\mathbb{P}_\beta^{U_\beta}$ is a complete subforcing of $\mathbb{P}_\beta$. The entries of $F(\beta)$ are required to be canonical lifts of $\mathbb{P}_\beta^{U_\beta}$ -names such that

$$\mathbb{P}_\beta^{U_\beta} \Vdash \text{“}\dot{x}_\beta, \dot{y}_\beta \in 2^\omega \text{ and } \dot{m}_\beta \in \omega\text{”}$$

Earlier names are identified with their canonical lifts to $\mathbb{P}_\beta$. Since $\eta_\beta \notin U_\beta$, the coordinate used for the new coding area does not occur in the data which determine the real coded at this stage. Let

$$\dot{Y}_\beta := \dot{Y}(\dot{x}_\beta, \dot{y}_\beta, \dot{m}_\beta, \dot{g}_{\eta_\beta}) \quad \text{and} \quad \dot{\mathbb{A}}_\beta := \mathbb{A}_D(\dot{Y}_\beta).$$

Then

$$\dot{\mathbb{R}}_{\beta+1} := \dot{\mathbb{R}}_\beta * \dot{\mathbb{A}}_\beta.$$

3. If $\lambda \leq \delta$ is a limit ordinal, then $\dot{\mathbb{R}}_\lambda$ is the finite-support limit of $\langle \dot{\mathbb{R}}_\beta \mid \beta < \lambda \rangle$.

The final forcing of the iteration is $\mathbb{P}_\delta$. A forcing is allowable if it is isomorphic to the final forcing of an allowable iteration over the reservoir.

Suppose that $\mathbb{P}$ and $\mathbb{Q}$ are allowable iterations over the reservoir of lengths $\delta \leq \varepsilon$, relative to $(F^\mathbb{P}, \vec{\eta}^\mathbb{P})$ and $(F^\mathbb{Q}, \vec{\eta}^\mathbb{Q})$, respectively. We say that $\mathbb{Q}$ is an allowable extension of $\mathbb{P}$, and write

$$\mathbb{Q} \triangleright \mathbb{P},$$

if

$$F^\mathbb{Q} \restriction \delta = F^\mathbb{P}, \quad \vec{\eta}^\mathbb{Q} \restriction \delta = \vec{\eta}^\mathbb{P},$$

and the iteration $\langle \dot{\mathbb{R}}_\beta^\mathbb{Q} \mid \beta \leq \varepsilon \rangle$ end-extends $\langle \dot{\mathbb{R}}_\beta^\mathbb{P} \mid \beta \leq \delta \rangle$.

Thus an allowable forcing has the schematic form

$$\left(\prod_{\eta < \omega_1}^{\text{cs}} \mathbb{C}_L(\omega_1) \right) * \left(\bigstar_{\beta < \delta}^{\text{fs}} \mathbb{A}_D(\dot{Y}_\beta) \right).$$

A condition may be represented by a pair (c, p) , where the product support of c is countable and the iteration support of p is finite. The first coordinate is the reservoir of coding areas; only the almost disjoint forcings are iterated. The injectivity of $\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle$ ensures that each coding area is used at most once. The requirement $\vec{\eta} \in L$ ensures that every branch inner model used in the localization arguments computes the same area assignment. At stage β , all bookkeeping names use only coordinates in U_β , while $\eta_\beta \notin U_\beta$ supplies the next coding area.

We identify a triple (x, y, m) with its fixed recursive real code. For a real u , $\gamma < \omega_1$, and $n < \omega$, put

$$\tau_u(\gamma, n) = \begin{cases} \omega\gamma + 2n + 1, & n \in u, \\ \omega\gamma + 2n, & n \notin u, \end{cases}$$

and, for $h \subseteq \omega_1$, let

$$\text{Sel}(h, u) := \{\tau_u(\gamma, n) : \gamma \in h, n < \omega\}.$$

Thus a stage which codes u on the area h uses precisely the branches $\langle b_\xi \mid \xi \in \text{Sel}(h, u) \rangle$.

Lemma 2.9. *Let $\mathbb{P}_\delta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\delta$ be an allowable forcing of length $\delta < \omega_1$.*

1. For every $\beta < \delta$,

$$\mathbb{P}_\beta \Vdash \text{“}\dot{\mathbb{A}}_\beta \text{ is Knaster and } |\dot{\mathbb{A}}_\beta| = \aleph_1\text{.”}$$

Consequently, $\mathbb{C}^{\text{area}} \Vdash \text{“}\dot{\mathbb{R}}_\delta \text{ is c.c.c. and has size at most } \aleph_1\text{.”}$

2. $\mathbb{P}_\delta$ preserves ω_1 and CH.

3. If $\mathbb{P}^0$ and $\mathbb{P}^1$ are allowable, then

$$\mathbb{P}^0 \times \mathbb{P}^1$$

is isomorphic to an allowable forcing. More precisely, after an L -definable reindexing of the two reservoir factors onto disjoint sets of coordinates, the product admits an allowable iteration over the global reservoir.

Proof. Clause (1) follows from Lemma 2.4 and the definition of $\mathbb{A}_D(Y)$. Indeed, D has size $\aleph_1$, so every almost disjoint coding forcing has size $\aleph_1$ and is Knaster. A finite-support iteration of c.c.c. forcings is c.c.c.; since $\delta < \omega_1$ and CH holds after the reservoir forcing, the iteration has size at most $\aleph_1$.

For (2), note first that $\mathbb{C}^{\text{area}}$, as computed in L , is σ -closed and has size $\aleph_1$. Since $W = L[H]$ is obtained from L by the c.c.c. branch product, Easton's lemma yields

$$W \models \text{"}\mathbb{C}^{\text{area}} \text{ is } \omega\text{-distributive."}$$

Hence the reservoir forcing preserves ω_1 , adds no reals, and preserves CH. In the reservoir extension, $\dot{\mathbb{R}}_\delta$ is a c.c.c. forcing of size at most $\aleph_1$ by (1). It therefore preserves ω_1 , and, since the ground model of this second step satisfies CH, it adds at most $\aleph_1$ many reals. Thus the final extension still satisfies CH.

For (3), write

$$\mathbb{P}^i = \mathbb{C}_i^{\text{area}} * \dot{\mathbb{R}}_{\delta_i}^i \quad (i < 2).$$

Fix in L a partition $\omega_1 = E_0 \dot{\cup} E_1$ and bijections $e_i : \omega_1 \rightarrow E_i$. Since the reservoirs are countable-support products of the same L -computed forcing,

$$\mathbb{C}_0^{\text{area}} \times \mathbb{C}_1^{\text{area}} \cong \mathbb{C}^{\text{area}}$$

via the reindexing e_0, e_1 . Under this isomorphism the coding areas used by the two factors are moved to disjoint sets of reservoir coordinates.

Over this common reservoir, the product

$$\dot{\mathbb{R}}_{\delta_0}^0 \times \dot{\mathbb{R}}_{\delta_1}^1$$

of the two finite-support almost-disjoint coding iterations is isomorphic to their concatenation: first use the translated factors from $\dot{\mathbb{R}}_{\delta_0}^0$ and then the translated factors from $\dot{\mathbb{R}}_{\delta_1}^1$. $\square$

Corollary 2.10. *Every allowable forcing preserves CH. More generally, suppose that*

$$\mathbb{P}_{\omega_1} = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_{\omega_1}$$

is an ω_1 -length allowable iteration over the reservoir whose restriction to every $\alpha < \omega_1$ is allowable. Then $\mathbb{P}_{\omega_1}$ preserves ω_1 and CH.

Proof. Only the second assertion requires comment. In the reservoir extension, $\dot{\mathbb{R}}_{\omega_1}$ is the finite-support iteration of the ω_1 many c.c.c. factors appearing in the iteration. It is c.c.c. and, by CH, has size $\aleph_1$. The argument from Lemma 2.9(2) now applies verbatim. $\square$

Let $K * G$ be $\mathbb{P}_\delta$ -generic over W . For $\beta < \delta$ set

$$u_\beta := \langle \dot{x}_\beta^{K*G}, \dot{y}_\beta^{K*G}, \dot{m}_\beta^{K*G} \rangle \quad \text{and} \quad h_\beta := \dot{h}_{\eta_\beta}^K.$$

We call $\{u_\beta : \beta < \delta\}$ the set of reals intentionally coded by the iteration.

Lemma 2.11. *Let*

$$\mathbb{P}_\delta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\delta$$

*be an allowable forcing over $W = L[H]$, and let $K * G$ be $\mathbb{P}_\delta$ -generic over W . If $x \in W[K][G] \cap 2^\omega$, then there are countable sets*

$$E_x \subseteq \omega_1, \quad I_x \subseteq \delta, \quad B_x \subseteq \omega_1$$

such that, putting

$$\text{Sel}_{I_x} := \bigcup_{\beta \in I_x} \text{Sel}(h_\beta, u_\beta),$$

we have

$$x \in L[H_{B_x \cup \text{Sel}_{I_x}}][K \restriction I_x][G \restriction I_x].$$

Here $G \restriction I_x$ denotes the generic sequence for the induced closed subiteration on I_x . Since $\vec{\eta} \in L$, the same area assignment belongs to every model $L[H_B]$ occurring in the localization argument; no branch support is needed to recover it. The set I_x may be chosen closed under the bookkeeping names of the iteration, and E_x may be chosen so that

$$\{\eta_\beta : \beta \in I_x\} \subseteq E_x.$$

Thus, whenever a stage belongs to I_x , the reservoir coordinate assigned to that stage also belongs to the localized reservoir support.

Proof. We prove the assertion by induction on the length δ of the allowable iteration. The statement is clear at length 0: the reservoir forcing adds no reals over W , and every real of $W = L[H]$ belongs to $L[H_B]$ for some countable $B \subseteq \omega_1$ by Theorem 2.7.

Assume the lemma is known for all shorter allowable iterations. Let $x \in W[K][G] \cap 2^\omega$, and fix a nice $\mathbb{P}_\delta$ -name $\dot{x}$ for x . A condition in $\mathbb{P}_\delta$ has the form $(c, \dot{p})$, where c is a condition in the reservoir product and $\dot{p}$ is a finite-support iteration condition. Let

$$I_0 = \bigcup \{\text{supp}_{\text{it}}(p) : p \text{ occurs in } \dot{x}\}$$

and let E_0 be the union of the reservoir supports of the conditions occurring in $\dot{x}$. Since $\dot{x}$ is a nice name for a real, $I_0 \subseteq \delta$ and $E_0 \subseteq \omega_1$ are countable.

We close I_0 under the names which define the iterands. Thus, whenever a stage β is retained, we also retain nice names for

$$\dot{x}_\beta, \quad \dot{y}_\beta, \quad \dot{m}_\beta,$$

and all iteration coordinates and reservoir coordinates occurring in those names. We also add η_β to the reservoir support. Repeating this operation only countably many times gives countable sets $I \subseteq \delta$ and $E \subseteq \omega_1$ such that

$$\{\eta_\beta : \beta \in I\} \subseteq E.$$

By construction, the name $\dot{x}$ and all names needed to define the iterands at coordinates in I are read from the induced closed subiteration on I together with the reservoir coordinates in E .

For each $\beta \in I$, the real u_β belongs to the extension by the initial iteration of length β . By the induction hypothesis, applied to that shorter iteration, there are countable sets $B_\beta \subseteq \omega_1$, $E_\beta \subseteq \omega_1$, and $I_\beta \subseteq \beta$ such that

$$u_\beta \in L[H_{B_\beta \cup \text{Sel}_{I_\beta}}][K \restriction E_\beta][G \restriction I_\beta].$$

Enlarge I and E by all I_β and E_β obtained in this way, and iterate the closure. Since only countably many stages are involved, the final sets, still denoted by I and E , are countable.

Let

$$B = \bigcup_{\beta \in I} B_\beta \cup B_0,$$

where B_0 is a countable branch support for the ground-model parameters which occur in the closed system of names. Again Theorem 2.7 gives such a B_0 , and B is countable. Put

$$\text{Sel}_I = \bigcup_{\beta \in I} \text{Sel}(h_\beta, u_\beta).$$

For every $\beta \in I$, the set Y_β used by the β -th almost disjoint coding factor is computed from the localized data for u_β , the reservoir coordinate g_{η_β} , and precisely the branches with indices in $\text{Sel}(h_\beta, u_\beta)$. Therefore the whole closed system of names, and in particular $\dot{x}$, is interpreted in

$$L[H_{B \cup \text{Sel}_I}][K \restriction E][G \restriction I].$$

Hence x belongs to this model. This proves the induction step and the lemma. $\square$

Lemma 2.12. *Let $\mathbb{P}_\delta$ be allowable, and let $K * G$ be generic over W . Then*

$$W[K][G] \models \Phi(u) \iff u = u_\beta \text{ for some } \beta < \delta.$$

The same conclusion holds for an ω_1 -length iteration over the reservoir all of whose proper initial segments are allowable.

Proof. If $u = u_\beta$, the real added by $\mathbb{A}_D(\dot{Y}_\beta)$ witnesses $\Psi(r, u)$ immediately after stage β . Since Ψ is Π_2^1 , this remains true in every later forcing extension by Shoenfield absoluteness. Hence every intentionally coded real satisfies Φ .

Conversely, work in $W[K][G]$ and suppose that r witnesses $\Phi(u)$. By the coding analysis of the preceding subsection, r decodes an unbounded coding area $h \subseteq \omega_1$, of the form

$$h = \{\varrho_L(a \restriction \xi) : \xi < \omega_1\}$$

for a function $a \in {}^{\omega_1}2$ all of whose proper initial segments belong to L , and

$$L[r] \models \text{“}S_{\tau_u(\gamma, n)} \text{ has an } \omega_1\text{-branch”} \quad (\gamma \in h, n < \omega).$$

We use the following elementary separation fact. If $a \neq a'$, then the corresponding areas $h(a)$ and $h(a')$ have bounded intersection. Indeed, after the first coordinate at which a and a' differ, no sufficiently long initial segment of one equals an initial segment of the other; injectivity of ϱ_L leaves only countably many common values. Hence the areas $\langle h_\beta \mid \beta < \delta \rangle$ are pairwise almost disjoint.

Assume towards a contradiction that $u \neq u_\beta$ for all $\beta < \delta$. Apply Lemma 2.11 to the two reals r and u , and take the union of the resulting supports. Thus there are countable sets

$$E \subseteq \omega_1, \quad I \subseteq \delta, \quad B \subseteq \omega_1$$

such that, with

$$\text{Sel}_I = \bigcup_{\beta \in I} \text{Sel}(h_\beta, u_\beta),$$

we have

$$r, u \in M := L[H_{B \cup \text{Sel}_I}][K \restriction E][G \restriction I].$$

We may assume that I is closed under the bookkeeping names of the allowable iteration.

We now choose a tree coordinate required by the alleged code for u but not present in the localized branch support. Put

$$B^{\text{blk}} := \{\gamma < \omega_1 : [\omega\gamma, \omega\gamma + \omega) \cap B \neq \emptyset\}.$$

This set is countable.

First suppose that $h \neq h_\beta$ for every $\beta \in I$. Since I is countable, choose

$$\gamma \in h \setminus \left(B^{\text{blk}} \cup \bigcup_{\beta \in I} (h \cap h_\beta) \right).$$

Then for any $n < \omega$,

$$\tau_u(\gamma, n) \notin B \cup \text{Sel}_I.$$

Fix $n = 0$ and set $\xi = \tau_u(\gamma, 0)$.

Now suppose that $h = h_{\beta_0}$ for some $\beta_0 \in I$. Since $u \neq u_{\beta_0}$, choose $n < \omega$ with $u(n) \neq u_{\beta_0}(n)$. Choose

$$\gamma \in h_{\beta_0} \setminus \left(B^{\text{blk}} \cup \bigcup_{\beta \in I \setminus \{\beta_0\}} (h_{\beta_0} \cap h_\beta) \right),$$

and put $\xi = \tau_u(\gamma, n)$. At the n -th pair of the γ -block, the stage β_0 selects the opposite tree, and no other stage in I uses this block. Hence again

$$\xi \notin B \cup \text{Sel}_I.$$

In either case, $\xi = \tau_u(\gamma, n)$ for some $\gamma \in h$ and $n < \omega$, while $\xi \notin B \cup \text{Sel}_I$. By Lemma 2.2, the tree S_ξ remains Suslin in $L[H_{B \cup \text{Sel}_I}]$. The reservoir part $K \restriction E$ is an L -computed product of ω_1 -Cohen forcings, and the almost disjoint part $G \restriction I$ is a finite-support iteration of Knaster forcings whose defining sets use only branches in $B \cup \text{Sel}_I$. Therefore the preservation facts from Lemma 2.4, together with the preservation of unused Suslin trees by the reservoir forcing, give

$$M \models "S_\xi \text{ is Suslin}."$$

But $r \in M$, and the displayed decoding property of r gives

$$L[r] \models "S_\xi \text{ has an } \omega_1\text{-branch}."$$

Since $L[r] \subseteq M$, this branch belongs to M , contradicting that $M \models "S_\xi \text{ is Suslin}"$. Hence $u = u_\beta$ for some $\beta < \delta$.

Finally, let the iteration have length ω_1 , and suppose that $\Phi(u)$ holds in the final extension, witnessed by r . Nice names for u and r involve only countably many conditions, and each condition has finite iteration support. Hence these names are supported by some $\alpha < \omega_1$. Since $\Psi(r, u)$ is Π_2^1 , Shoenfield absoluteness implies that it already holds after the α -th initial segment. Applying the first part of the lemma to that allowable initial segment shows that u was intentionally coded before α . $\square$

2.5 Motivation for the allowability hierarchy

Given the notion of allowable forcing, we now form successive approximations to the class of forcings which we eventually want to use. We first describe the idea leading to these approximations.

Fix a recursive enumeration

$$\langle \varphi_m(v_0, v_1) \mid m \in \omega \rangle$$

of the Π_3^1 -formulas with two free real variables, and put

$$A_m := \{(x, y) \in (2^\omega)^2 : \varphi_m(x, y)\}.$$

We write f_m for the uniformizing function which we intend to define for A_m , and write $f(m, x)$ for $f_m(x)$. The goal is to choose, for every $m \in \omega$ and every real x for which

$$A_{m,x} := \{y \in 2^\omega : (x, y) \in A_m\}$$

is nonempty, a real $f(m, x)$ such that

$$(x, f(m, x)) \in A_m,$$

and to code every triple (x, y, m) such that

$$y \neq f(m, x) \quad \text{and} \quad (x, y) \in A_m$$

into $\vec{S}$. The final forcing will have length ω_1 , and each of its countable initial segments will be allowable. Since the statement

$$“(x, y, m) \text{ is coded into } \vec{S}”$$

is Σ_3^1 , the requirement

$$(x, y) \in A_m \quad \text{and} \quad (x, y, m) \text{ is not coded into } \vec{S}$$

is Π_3^1 . The bookkeeping of the final iteration will ensure that, whenever $A_{m,x} \neq \emptyset$, exactly one such y remains.

The idea behind the definition can be seen in the following toy example. Work in W , let $x \in W$, and suppose that

$$A_{m,x} = \{y_0, y_1\}.$$

Assume in addition that no allowable forcing over W adds a new member to $A_{m,x}$. We consider only the problem of defining $f(m, x)$ and omit, for the moment, the interaction with the other sections.

Call $y \in A_{m,x}$ *eliminable* if there is an allowable forcing $\mathbb{P}$ such that

$$\mathbb{P} \Vdash (x, y) \notin A_m.$$

Call y *persistent* if it is not eliminable. Thus, if y_0 is persistent, we may reserve y_0 as the value of $f(m, x)$ and code (x, y_1, m) into $\vec{S}$. If y_0 is eliminable, witnessed by an allowable forcing $\mathbb{P}$, we consider y_1 . If y_1 is persistent, we reserve y_1 and code (x, y_0, m) .

It remains to consider the case in which both values are eliminable. Let $\mathbb{P}$ and $\mathbb{Q}$ witness, respectively, that

$$\mathbb{P} \Vdash (x, y_0) \notin A_m \quad \text{and} \quad \mathbb{Q} \Vdash (x, y_1) \notin A_m.$$

By Lemma 2.9, the product

$$\mathbb{P} \times \mathbb{Q}$$

is isomorphic to an allowable forcing. Identifying the product with its normalized allowable presentation, upward absoluteness gives

$$\mathbb{P} \times \mathbb{Q} \Vdash (x, y_0) \notin A_m \wedge (x, y_1) \notin A_m.$$

By the assumption that no new member is added to the section, the product forces

$$A_{m,x} = \emptyset.$$

In this case there is no value to uniformize at x .

This argument isolates the two operations which are needed later:

reserve a value which is persistent under the forcings under consideration;
combine eliminating forcings by taking products.

(1)

The difficulty is that an eliminating forcing may add codes which interfere with other sections. Thus it is not enough to use arbitrary allowable forcings as witnesses. We need a class $\mathcal{P}$ of allowable forcings such that the following rule produces forcings in $\mathcal{P}$ again. For every task (x, m) considered by the bookkeeping, either

$$\exists y \in A_{m,x} \forall \mathbb{Q} \in \mathcal{P} (\mathbb{Q} \text{ extending the current forcing} \rightarrow \mathbb{Q} \Vdash (x, y) \in A_m),$$

in which case one such y is reserved and the other values are coded, or every value under consideration can be eliminated by a forcing in $\mathcal{P}$.

This turns the construction into a fixed point problem. Let

$$\mathcal{A}_0^W$$

be the class of allowable iterations. The first refinement searches for values which are persistent under $\mathcal{A}_0^W$ -extensions; the resulting iterations form $\mathcal{A}_1^W$. The next refinement searches also for values which are persistent under $\mathcal{A}_1^W$ -extensions, and so on. We obtain a decreasing hierarchy

$$\mathcal{A}_0^W \supseteq \mathcal{A}_1^W \supseteq \cdots \supseteq \mathcal{A}_\alpha^W \supseteq \cdots$$

The case $\alpha = 1$ is the first approximation described above.

The reason for continuing the construction is the usual moving-target problem. In the definition of 1-allowability we test persistence under 0-allowable extensions, but the iteration which we produce is itself 1-allowable. We should therefore also test persistence under 1-allowable extensions. Repeating this argument leads to the transfinite hierarchy defined in the next subsection.

2.6 The hierarchy of α -allowable iterations

We formulate the definition for iterations over the fixed reservoir. A forcing is called α -allowable if it is isomorphic to the final forcing of an α -allowable iteration.

We use the canonical $<_L$ -coding convention for objects in $H(\omega_2)^W$. Every object used below is represented by its $<_L$ -least Br-name in L , and codes are compared by $<_L$.

We also fix a support convention which takes account of both parts of an allowable iteration. For $E \subseteq \omega_1$, put

$$\mathbb{C}_E^{\text{area}} := \prod_{\eta \in E}^{\text{cs}} \mathbb{C}_L(\omega_1).$$

Suppose that

$$\mathbb{P}_\beta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\beta$$

is an initial segment of an allowable iteration relative to $(F, \vec{\eta})$, and put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

Let $\vec{\tau}$ be a finite tuple of $\mathbb{P}_\beta$ -names for reals or natural numbers. A pair

$$(E, A) \in [U_\beta]^{\leq \omega} \times [\beta]^{\leq \omega}$$

is *admissible for $\vec{\tau}$* if the induced iteration on the stages in A , using only the reservoir coordinates in E , is recursively defined, its final forcing

$$\mathbb{P}_{E,A} := \mathbb{C}_E^{\text{area}} * \dot{\mathbb{R}}_{E,A}$$

is a complete subforcing of $\mathbb{P}_\beta$, and every member of $\vec{\tau}$ is the canonical lift of a $\mathbb{P}_{E,A}$ -name. In particular, the pair is closed under the names defining the retained iterands: if $\gamma \in A$, then the coding coordinate η_γ belongs to E , and the names which define the γ -th iterand are already names over the induced iteration on $A \cap \gamma$ and the reservoir coordinates in E .

The reservoir support E is countable and is contained in U_β . Its role is to record the coordinates which occur in the names defining the relevant part of the almost disjoint iteration. It does not record new real information. Indeed, the reservoir and each of its coordinate factors are L -computed σ -closed forcings. After the c.c.c. forcing which produces W and the c.c.c.

almost disjoint tail below β , Easton's lemma implies that the unused reservoir factor on $\omega_1 \setminus U_\beta$ is ω -distributive. Thus every real in the full intermediate extension has an equivalent name which uses only coordinates in U_β . The same observation applies after a permutation of the reservoir coordinates. Combining this with the usual nice-name argument for the c.c.c. tail, and then closing recursively under the names defining the retained iterands, shows that every finite tuple of reals and natural numbers has a tuple of equivalent names with an admissible support pair whose two coordinates are countable.

We put

$$\text{supp}_{\text{AP}}(\vec{\tau}, \mathbb{P}_\beta) := \text{the } <_L \text{-least pair } (E, A) \text{ which is admissible for } \vec{\tau}.$$

The iteration support A need not be an ordinal.

For an admissible support pair $\mathfrak{a} = (E, A)$ occurring in this sense, put

$$\mathbb{P}_{\mathfrak{a}} := \mathbb{P}_{E,A}.$$

Whenever an ambient generic $K * G_\beta$ is fixed, write

$$G_{\mathfrak{a}} := K_E * G_A, \quad M_{\mathfrak{a}} := W[K_E][G_A].$$

If $\mathfrak{b} = (F, B)$ and $\mathfrak{a} = (E, A)$ are admissible support pairs, $F \subseteq E$, $B \subseteq A$, and $\mathbb{P}_{\mathfrak{b}}$ is a complete subforcing of $\mathbb{P}_{\mathfrak{a}}$, we write

$$\mathfrak{b} \preccurlyeq \mathfrak{a}.$$

For the persistence relation we distinguish literal end-extensions from complete subiterations obtained after reindexing the reservoir. The notation

$$\mathbb{Q} \triangleright \mathbb{P}$$

continues to mean that $\mathbb{Q}$ literally end-extends $\mathbb{P}$ over the same reservoir, as in Definition 2.8.

We introduce a second relation which is invariant under the normalization of products. A *reservoir block* is a pair (B, e) such that

$$B \in L, \quad B \subseteq \omega_1, \quad |B| = \omega_1,$$

and

$$e : \omega_1 \longrightarrow B$$

is a bijection belonging to L . If $\mathbb{P}$ is an allowable iteration over the reservoir, let $e_*\mathbb{P}$ denote its reindexed copy obtained by replacing

$$\mathbb{C}^{\text{area}} \quad \text{with} \quad \mathbb{C}_B^{\text{area}},$$

replacing every canonical coordinate name $\dot{g}_\eta$ by $\dot{g}_{e(\eta)}$, and translating recursively all names in the finite-support tail.

Let $\mathbb{P}$ and $\mathbb{Q}$ be allowable iterations of lengths δ and ε , respectively. We write

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P}$$

and say that $\mathbb{Q}$ is a *complete reservoir extension* of $\mathbb{P}$ if there are a reservoir block (B, e) and a set $A \in W$, $A \subseteq \varepsilon$, of order type δ such that the following hold. Writing

$$\pi : \delta \longrightarrow A$$

for the increasing enumeration of A , the stages in A are closed under the names defining their iterands when only the reservoir coordinates in B are retained. Hence the induced iteration

$$\mathbb{Q}_{B,A} := \mathbb{C}_B^{\text{area}} * \dot{\mathbb{R}}_{B,A}^{\mathbb{Q}}$$

is recursively defined and is a complete subforcing of $\mathbb{Q}$. Moreover, reindexing the reservoir by e and the stages by π induces an isomorphism

$$\iota_{e,\pi} : \mathbb{P} \cong \mathbb{Q}_{B,A}.$$

The reservoir block, the set of stages, and the isomorphism are regarded as part of the witness to $\mathbb{Q} \succeq_{\text{res}} \mathbb{P}$.

The relation $\succeq_{\text{res}}$ is reflexive and transitive. Reflexivity is witnessed by the identity reindexing and the full set of stages. For transitivity, compose the two reservoir reindexings and the two increasing maps between the retained stages; complete subforcings compose. In particular,

$$\mathbb{Q} \triangleright \mathbb{P} \implies \mathbb{Q} \succeq_{\text{res}} \mathbb{P}.$$

Further, if an allowable iteration is the normalized reservoir presentation of

$$\mathbb{P}^0 \times \mathbb{P}^1,$$

obtained by moving the two copies of the reservoir to disjoint blocks of coordinates and concatenating their almost disjoint tails, then each factor occurs in this iteration as a complete reservoir subiteration.

If G is $\mathbb{P}$ -generic and

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P}$$

is witnessed by $\iota_{e,\pi}$, we identify G with its image under this isomorphism and write

$$\mathbb{Q}/G$$

for the quotient forcing over the chosen complete reservoir subiteration.

We need a way how to single out reals which should become the values of our uniformizing functions. Suppose we are given a real x and a Π_3^1 set A_m and we know that there is a real y such that (x, y) is persistent in A_m , that is it will remain in A_m in all further allowable extensions. The task is to

determine which real z to pick as a value for $f(m, x)$. A first naive attempt is to simply pick the $<_L$ -least name σ for such a real and let $z = \sigma^G$, for G the generic filter, but this does not work for technical reasons.

How to do this is described now. Let $\mathfrak{a} = (E, A)$ be an admissible support pair and let $z \in M_{\mathfrak{a}}$ be a real, or a fixed finite tuple of reals and natural numbers. In the application $z = \langle x, y, m \rangle$. A *presentation of z below $\mathfrak{a}$* is a quadruple

$$c = (\mathfrak{b}, \mathbb{R}, p, \dot{\sigma})$$

such that

$$\mathfrak{b} \preceq \mathfrak{a}, \quad z \in M_{\mathfrak{b}},$$

$\mathbb{R} \in M_{\mathfrak{b}}$ is allowable over $M_{\mathfrak{b}}$, the coding areas used by $\mathbb{R}$ are disjoint from those already used by $\mathfrak{b}$, $p \in \mathbb{R}$, and

$$M_{\mathfrak{b}} \models p \Vdash_{\mathbb{R}} \dot{\sigma} = \check{z}.$$

Using the fixed coding convention, c determines the pointed allowable presentation

$$d_c = (\mathbb{P}_{\mathfrak{b}} * \mathbb{R}, (1_{\mathbb{P}_{\mathfrak{b}}}, \dot{p}), \dot{\sigma}),$$

where $\dot{\mathbb{R}}$ and $\dot{p}$ are the canonical $\mathbb{P}_{\mathfrak{b}}$ -names for $\mathbb{R}$ and p , and $\dot{\sigma}$ is lifted to the two-step forcing.

For pointed allowable presentations

$$d = (\mathbb{S}, q, \dot{\tau}), \quad e = (\mathbb{T}, r, \dot{\sigma}),$$

let $\equiv_{\text{pres}}$ be the equivalence relation generated by the following relation: d and e are related if there are a normalized allowable forcing $\mathbb{U}$, allowable complete embeddings

$$i : \mathbb{S} \longrightarrow \mathbb{U}, \quad j : \mathbb{T} \longrightarrow \mathbb{U},$$

and a condition $u \in \mathbb{U}$ such that

$$u \leq i(q), j(r) \quad \text{and} \quad u \Vdash_{\mathbb{U}} i(\dot{\tau}) = j(\dot{\sigma}).$$

Here allowable complete embedding means that the quotient forcing is allowable. Define

$$\text{pcode}(d) := \min_{<_L} \{\text{code}(e) : e \equiv_{\text{pres}} d\}.$$

Thus $\text{pcode}(d)$ is the canonical code of the presentation class of d . The definition is made in the ground model from the fixed coding of allowable forcings, embeddings, conditions, names, and the forcing relation.

For $z \in M_{\mathfrak{a}}$, we define the orbit of z relative to $\mathfrak{a}$ to be

$$\mathcal{O}_{\mathfrak{a}}(z) := \{\text{pcode}(d_c) : c \text{ is a presentation of } z \text{ below } \mathfrak{a}\}$$

and

$$\text{key}_{\mathfrak{a}}(z) := \min_{<_L} \mathcal{O}_{\mathfrak{a}}(z).$$

The orbit is nonempty, since we may take $\mathfrak{b} = \mathfrak{a}$ and $\mathbb{R}$ trivial. It remembers only the allowable presentation class of z , not the support or factorization in which z was first seen.

Lemma 2.13. *Suppose that*

$$\mathfrak{b} \preccurlyeq \mathfrak{a} \quad \text{and} \quad z \in M_{\mathfrak{b}}.$$

Then

$$\mathcal{O}_{\mathfrak{b}}(z) = \mathcal{O}_{\mathfrak{a}}(z),$$

and therefore

$$\text{key}_{\mathfrak{b}}(z) = \text{key}_{\mathfrak{a}}(z).$$

Proof. The inclusion $\mathcal{O}_{\mathfrak{b}}(z) \subseteq \mathcal{O}_{\mathfrak{a}}(z)$ is immediate. Conversely, let

$$c = (\mathfrak{c}, \mathbb{R}, p, \dot{\sigma})$$

be a presentation of z below $\mathfrak{a}$. Let $\mathfrak{d}$ be the admissible closure of $\mathfrak{b} \cup \mathfrak{c}$ inside $\mathfrak{a}$. Since $z \in M_{\mathfrak{b}} \cap M_{\mathfrak{c}}$, the lifted $\mathfrak{b}$ -presentation of z and the lifted $\mathfrak{c}$ -presentation of z have the same value in $M_{\mathfrak{d}}$. By the truth lemma, after strengthening in the quotient to $\mathfrak{d}$, this equality is forced.

Thus d_c and the presentation obtained from $\mathfrak{b}$, followed by the quotient to $\mathfrak{d}$ and then by $\mathbb{R}$, have a common allowable refinement below which their names are forced to be equal. They therefore determine the same presentation class. Hence $\text{pcode}(d_c) \in \mathcal{O}_{\mathfrak{b}}(z)$. This proves the reverse inclusion, and the equality of keys follows by taking the $<_L$ -least element of the common orbit. $\square$

At a stage with local admissible support $\mathfrak{a}$ and given a pair (x, m) , where x is a real and $m \in \omega$, we first minimize the persistence rank. Among the reals of least rank we compare

$$\text{key}_{\mathfrak{a}}(\langle x, y, m \rangle),$$

and choose the real with $<_L$ -least key. This is the canonical selection rule for the value of $f(m, x)$.

We have assembled all the relevant notions to define the main concept of this work. We define ρ -allowability by induction on ρ .

Definition 2.14. *A 0-allowable iteration is an allowable iteration over the reservoir in the sense of Definition 2.8. For purposes of the hierarchy, it is paired with the canonical auxiliary name for the empty set.*

Let $0 < \rho$, and assume that ζ -allowability has been defined for every $\zeta < \rho$. Let $\delta < \omega_1$, let

$$F = \langle (\dot{x}_{\beta}, \dot{z}_{\beta}, \dot{m}_{\beta}) \mid \beta < \delta \rangle \in W$$

be a bookkeeping function, and let

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle \in L$$

be an injective sequence of ordinals below ω_1 . For $\beta \leq \delta$, put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

Here $\dot{z}_\beta$ is the bookkeeping guess for a value in the $\dot{x}_\beta$ -section of $A_{\dot{m}_\beta}$. We require that, whenever $\mathbb{P}_\beta$ has been defined and

$$(E_\beta, A_\beta) := \text{supp}_{\text{AP}}(F(\beta), \mathbb{P}_\beta),$$

then

$$\mathbb{P}_{E_\beta, A_\beta} \Vdash \text{“}\dot{x}_\beta, \dot{z}_\beta \in 2^\omega \text{ and } \dot{m}_\beta \in \omega\text{”}$$

and

$$E_\beta \subseteq U_\beta.$$

Thus the local bookkeeping data at stage β use only coding-area coordinates assigned before stage β . Since $\vec{\eta}$ is injective,

$$\eta_\beta \notin U_\beta,$$

and therefore $\eta_\beta \notin E_\beta$. The entire assignment belongs to L , so it is computed in the same way in every branch inner model of W . By Easton's lemma, requiring stage-respecting names does not remove any real which can occur in the construction. We shall consider only pairs $(F, \vec{\eta})$ satisfying these requirements.

A ρ -allowable iteration relative to $(F, \vec{\eta})$ is a sequence

$$\mathbb{P} = \langle \mathbb{P}_\beta \mid \beta \leq \delta \rangle, \quad \mathbb{P}_\beta = \mathbb{C}^{\text{area}} * \dot{\mathbb{R}}_\beta,$$

together with auxiliary names $\langle \dot{I}_\beta \mid \beta \leq \delta \rangle$, defined as follows.

At stage 0,

$$\dot{\mathbb{R}}_0 = \mathbf{1} \quad \text{and} \quad \dot{I}_0 = \emptyset.$$

Suppose that $\beta < \delta$, that $\mathbb{P}_\beta$ and $\dot{I}_\beta$ have been defined, and let

$$K * G_\beta$$

be $\mathbb{P}_\beta$ -generic over W . Put

$$E := E_\beta, \quad A := A_\beta, \quad K_E := K \restriction E, \quad G_A := G_\beta \restriction A,$$

so that

$$G_{E,A} := K_E * G_A$$

is $\mathbb{P}_{E,A}$ -generic over W . Let

$$x := \dot{x}_\beta^{G_{E,A}}, \quad z := \dot{z}_\beta^{G_{E,A}}, \quad m := \dot{m}_\beta^{G_{E,A}}.$$

The coding area assigned to stage β is the reservoir coordinate η_β . Since

$$E \subseteq U_\beta \quad \text{and} \quad \eta_\beta \notin U_\beta,$$

the area used at this stage is outside the local reservoir support. Earlier names are identified with their canonical lifts.

Put

$$\mathfrak{a}_\beta := (E, A).$$

For $\zeta < \rho$ and $y \in M_{\mathfrak{a}_\beta} \cap 2^\omega$, say that y is ζ -persistent for (x, m) over $\mathbb{P}_\beta$ if

$$W[K][G_\beta] \models (x, y) \in A_m$$

and, for every ζ -allowable iteration $\mathbb{Q} \in W$ together with a fixed witness to

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P}_\beta,$$

we have

$$W[K][G_\beta] \models \text{“}\mathbb{Q}/(K * G_\beta) \Vdash (x, y) \in A_m\text{”}.$$

Equivalently, using the standing convention of restricting a forcing below a condition, no ζ -allowable complete reservoir extension of $\mathbb{P}_\beta$ forces

$$(x, y) \notin A_m.$$

Let

$$C_\beta^\rho(K * G_\beta) := \{(\zeta, y) \in \rho \times (W[K_E][G_A] \cap 2^\omega) : y \text{ is } \zeta\text{-persistent for } (x, m) \text{ over } \mathbb{P}_\beta\}.$$

We distinguish two cases.

(a) The first uniformization case. Assume that $C_\beta^\rho(K * G_\beta) \neq \emptyset$. Choose

$$\zeta_\beta := \min\{\zeta : \exists y ((\zeta, y) \in C_\beta^\rho(K * G_\beta))\},$$

and then choose y_β so that $(\zeta_\beta, y_\beta) \in C_\beta^\rho(K * G_\beta)$ and

$$\text{key}_{\mathfrak{a}_\beta}(\langle x, y_\beta, m \rangle)$$

is $<_L$ -least among the keys of all such reals. Let $z_\beta^* = z$ if $z \neq y_\beta$. If $z = y_\beta$, let z_β^* be the real $w \neq y_\beta$ in $W[K_E][G_A]$ for which

$$\text{key}_{\mathfrak{a}_\beta}(\langle x, w, m \rangle)$$

is $<_L$ -least.

The next almost disjoint coding factor is

$$\dot{\mathbb{A}}_\beta := \mathbb{A}_D(\dot{Y}(\dot{x}_\beta, \dot{z}_\beta^*, \dot{m}_\beta, \dot{g}_{\eta_\beta})),$$

where $\dot{z}_\beta^*$ is the name obtained from the preceding uniform choice. We put

$$\dot{\mathbb{R}}_{\beta+1} := \dot{\mathbb{R}}_\beta * \dot{\mathbb{A}}_\beta$$

and, for $K * G_{\beta+1}$ extending $K * G_\beta$, define

$$\dot{I}_{\beta+1}^{K * G_{\beta+1}} := \dot{I}_\beta^{K * G_\beta} \cup \{(x, y_\beta, m, \zeta_\beta)\}.$$

In this case y_β is declared to be a tentative value for $f_m(x)$ of rank ζ_β . The triple (x, z_β^*, m) is coded instead.

(b) The second uniformization case. Assume that $C_\beta^\rho(K * G_\beta) = \emptyset$. In this case no real in $M_{\mathfrak{a}_\beta} \cap 2^\omega$ passes the test at any rank below ρ , and the construction uses the bookkeeping guess. We put

$$\dot{\mathbb{A}}_\beta := \mathbb{A}_D(\dot{Y}(\dot{x}_\beta, \dot{z}_\beta, \dot{m}_\beta, \dot{g}_{\eta_\beta})),$$

$$\dot{\mathbb{R}}_{\beta+1} := \dot{\mathbb{R}}_\beta * \dot{\mathbb{A}}_\beta,$$

and

$$\dot{I}_{\beta+1}^{K * G_{\beta+1}} := \dot{I}_\beta^{K * G_\beta}.$$

Thus no new tentative value is added at stage β .

If $\lambda \leq \delta$ is a limit ordinal, let $\dot{\mathbb{R}}_\lambda$ be the finite-support limit of

$$\langle \dot{\mathbb{R}}_\beta \mid \beta < \lambda \rangle$$

and define, for $K * G_\lambda$ generic,

$$\dot{I}_\lambda^{K * G_\lambda} := \bigcup_{\beta < \lambda} \dot{I}_\beta^{K * G_\beta}.$$

All choices above are made from the fixed order $<_L$ on codes and are uniform in $K * G_\beta$. Hence they determine forcing names in W . The notation $K * G_\beta$ is only semantic shorthand; no generic filter is chosen during the recursion.

If the recursion produces $(\mathbb{P}_\delta, \dot{I}_\delta)$, we say that this pair is ρ -allowable relative to $(F, \vec{\eta})$. We call $\mathbb{P}_\delta$ ρ -allowable if it is ρ -allowable relative to some bookkeeping function F and some injective area assignment $\vec{\eta} \in L$ satisfying the displayed requirements.

For a generic filter $K * G$ for the final forcing, we write

$$I^{K * G} := \dot{I}_\delta^{K * G}.$$

The interpreted auxiliary set $I^{K*G} \in W[K][G]$ is the set of tentative values for the uniformizing functions. If $(x, y, m, \xi) \in I^{K*G}$, then y is a tentative value for $f_m(x)$ of rank ξ . The same value may occur with more than one rank. The auxiliary set does not enter the definition of the forcing factors; it records the tentative values selected by the recursion.

The case $\rho = 1$ is the notion of 1-allowability. Since the only ordinal below 1 is 0, clause (a) asks whether there is a value which is persistent under all allowable, that is, 0-allowable, continuations. Thus Definition 2.14 specializes to the intended notion of 1-allowability when $\rho = 1$.

The restriction to reals in

$$W[K_E][G_A]$$

is part of the definition. We do not scan all reals in $W[K][G_\beta]$. The reservoir forcing itself adds no reals, by Easton's lemma, but the coordinate set E is still needed to record the parameters on which the induced almost disjoint iteration depends. This support stratification is used in the product argument: after the reservoir coordinates and iteration stages of two factors are moved to disjoint blocks, the names belonging to one factor are evaluated in the corresponding translated complete subiteration.

At a successor level $\rho + 1$, the definition permits the additional rank ρ and therefore tests persistence under ρ -allowable continuations. At a limit λ , it tests all ranks $\zeta < \lambda$ at once. No separate limit-stage definition of allowability is required.

For formulas whose relevant instances are preserved by every forcing under consideration, clause (a) must apply whenever the section is nonempty. Thus tentative values already occur at the first derivative. We will not need a separate assertion that the inclusions between consecutive classes are proper.

Lemma 2.15. *Let $\xi < \alpha$. Every α -allowable forcing is ξ -allowable. Consequently,*

$$\mathcal{A}_0^W \supseteq \mathcal{A}_1^W \supseteq \cdots \supseteq \mathcal{A}_\alpha^W \supseteq \cdots,$$

where $\mathcal{A}_\alpha^W$ denotes the class of α -allowable forcings over W .

Proof. The case $\xi = 0$ is immediate, since 0-allowability is just allowability. Let $0 < \xi < \alpha$, and let $(\mathbb{P}, \dot{I})$ be α -allowable relative to $(F, \vec{\eta})$. We define, by recursion on the stages, a bookkeeping function F' such that the ξ -construction relative to $(F', \vec{\eta})$ produces the same forcing iteration.

Assume that the two constructions agree below a stage β . Let $\mathfrak{a}_\beta = (E_\beta, A_\beta)$ be the local admissible support used by the α -construction at this stage, and let the corresponding local data be (x, z, m) .

Suppose first that the α -construction is in clause (a) and selects a real y_β of rank $\rho < \xi$. For every $\zeta < \xi$, the persistence test is the same in the two constructions, since it only refers to ζ -allowable complete reservoir extensions

of the common initial forcing. The local support is the same, and the key

$$\text{key}_{\mathfrak{a}_\beta}(\langle x, y, m \rangle)$$

is computed from the same orbit. Hence the two constructions have the same least rank below ξ and then choose the same real by the $<_L$ -least key. We put $F'(\beta) = F(\beta)$, and the same coding factor is used.

Suppose next that the α -construction is in clause (a) but selects a real of rank $\rho \geq \xi$. Then no real in $M_{\mathfrak{a}_\beta}$ passes the persistence test at any rank below ξ . Thus the ξ -construction must be in clause (b). We choose $F'(\beta)$ to be a stage-respecting name, over the same local support, for the triple which the α -construction codes at stage β . Clause (b) of the ξ -construction therefore adds exactly the same almost disjoint coding factor.

Finally, if the α -construction is already in clause (b), then no real passes the test at any rank below α , hence none passes at a rank below ξ . We put $F'(\beta) = F(\beta)$, so the ξ -construction is also in clause (b) and uses the same factor.

The choices made in the recursion are uniform in the generic filter. The new bookkeeping names are chosen in the local models used by the α -construction, whose reservoir supports are contained in the corresponding U_β . Hence $(F', \vec{\eta})$ satisfies the stage-respecting support requirement. At limit stages both constructions take the same finite-support limit. Therefore the final forcing of the given α -allowable iteration is also ξ -allowable. $\square$

Lemma 2.16. *Let α be an ordinal. If $\mathbb{P}^0$ and $\mathbb{P}^1$ are α -allowable, then*

$$\mathbb{P}^0 \times \mathbb{P}^1$$

is isomorphic to an α -allowable forcing. More precisely, a normalized α -allowable presentation of the product is obtained from the two given presentations by an L -definable reindexing of the two reservoirs onto disjoint blocks and by concatenating the translated almost disjoint tails. The resulting bookkeeping function belongs to W , and the resulting injective area assignment belongs to L .

Proof. We argue by induction on α , proving the normalized form stated above. For $\alpha = 0$, this is exactly Lemma 2.9(3).

Assume that the assertion holds for every $\xi < \alpha$. Let $(F_i, \vec{\eta}^i)$ witness the α -allowability of $\mathbb{P}^i$, and let δ_i be the length of $\mathbb{P}^i$ ($i < 2$). Fix in L a partition

$$\omega_1 = B_0 \dot{\cup} B_1$$

and bijections

$$e_i : \omega_1 \longrightarrow B_i \quad (i < 2).$$

Move the reservoir of $\mathbb{P}^i$ to B_i by e_i . We present the product by first taking the translated almost disjoint tail of $\mathbb{P}^0$ and then the translated tail of $\mathbb{P}^1$. Let

$$\mathbb{S} = \langle \mathbb{S}_\gamma \mid \gamma \leq \delta_0 + \delta_1 \rangle$$

be this normalized product presentation. Its final forcing is isomorphic to $\mathbb{P}^0 \times \mathbb{P}^1$. Put

$$\vec{\eta}^\otimes = \langle e_0(\eta_\beta^0) \mid \beta < \delta_0 \rangle \frown \langle e_1(\eta_\beta^1) \mid \beta < \delta_1 \rangle.$$

Then $\vec{\eta}^\otimes \in L$ is injective. Let $F^\otimes$ be obtained by translating F_0 and F_1 by e_0 and e_1 , respectively, and concatenating the translated sequences. Then $F^\otimes \in W$.

It remains to check the stage rule. The first block is handled by the same argument as below, with only the reindexing e_0 , so we consider a stage $\delta_0 + \beta$ in the second block. Let $\beta < \delta_1$, and let

$$\mathfrak{a}_\beta^1 = (E, A)$$

be the local admissible support of the corresponding stage in the construction of $\mathbb{P}^1$. In the product presentation the translated support is

$$\mathfrak{a}_{\delta_0+\beta}^\otimes := (e_1[E], \delta_0 + A), \quad \delta_0 + A = \{\delta_0 + \gamma : \gamma \in A\}.$$

It is admissible, since

$$E \subseteq \text{ran}(\vec{\eta}^1 \restriction \beta)$$

implies

$$e_1[E] \subseteq \text{ran}(\vec{\eta}^\otimes \restriction (\delta_0 + \beta)).$$

The local models

$$W[K \restriction e_1[E]][G \restriction (\delta_0 + A)] \quad \text{and} \quad W[K^1 \restriction E][G^1 \restriction A]$$

are canonically isomorphic. We identify their reals through this isomorphism.

Fix $\xi < \alpha$ and a local real y . We claim that y is ξ -persistent for (x, m) at the stage β of $\mathbb{P}^1$ iff its translated copy is ξ -persistent at the stage $\delta_0 + \beta$ of $\mathbb{S}$. If y is not persistent in $\mathbb{P}^1$, let $\mathbb{Q}$ be a ξ -allowable complete reservoir extension of $\mathbb{P}_\beta^1$ whose quotient forces

$$(x, y) \notin A_m.$$

By Lemma 2.15, $\mathbb{P}^0$ is ξ -allowable, and by the induction hypothesis $\mathbb{P}^0 \times \mathbb{Q}$ has a normalized ξ -allowable presentation. This presentation contains $\mathbb{S}_{\delta_0+\beta}$ as a complete reservoir subiteration, and its quotient still forces

$$(x, y) \notin A_m.$$

Hence the translated y is not persistent in the product presentation.

Conversely, if a ξ -allowable complete reservoir extension of $\mathbb{S}_{\delta_0+\beta}$ forces

$$(x, y) \notin A_m,$$

then, since $\mathbb{S}_{\delta_0+\beta}$ contains the translated copy of $\mathbb{P}_\beta^1$ as a complete reservoir subiteration, the same forcing is also a ξ -allowable complete reservoir extension of $\mathbb{P}_\beta^1$ by transitivity of $\succeq_{\text{res}}$. Thus y is not persistent in the original construction of $\mathbb{P}^1$. This proves the claim.

The key comparison is also preserved. The reindexing e_1 sends presentations below $\mathfrak{a}_\beta^1$ bijectively to presentations below $\mathfrak{a}_{\delta_0+\beta}^\otimes$, and the corresponding pointed presentations have the same presentation class. Hence, for every local real y ,

$$\text{key}_{\mathfrak{a}_\beta^1}(\langle x, y, m \rangle) = \text{key}_{\mathfrak{a}_{\delta_0+\beta}^\otimes}(\langle x, y, m \rangle).$$

Equivalently, the two constructions compute the same orbits and therefore the same $<_L$ -least keys. Thus they have the same least persistence rank and, at that rank, choose the same real. If the selected value equals the bookkeeping value, the same key comparison also chooses the same real to be coded instead. Therefore the translated stage of $\mathbb{S}$ uses exactly the translated coding factor from $\mathbb{P}^1$.

The same verification applies to every stage of the second block, and the first block was already covered by the reindexing argument. At limits the finite-support limits agree. Hence $\mathbb{S}$ is an α -allowable iteration relative to $(F^\otimes, \vec{\eta}^\otimes)$. Since the final forcing of $\mathbb{S}$ is isomorphic to $\mathbb{P}^0 \times \mathbb{P}^1$, the product is isomorphic to an α -allowable forcing. $\square$

Lemma 2.17. *Let $(\mathbb{P}, \dot{I})$ be α -allowable, let G be $\mathbb{P}$ -generic over W , and suppose that*

$$(x, y, m, \xi) \in \dot{I}^G \quad \text{for some } \xi < \alpha.$$

Then

$$W[G] \models (x, y) \in A_m.$$

Moreover, if $\mathbb{Q}$ is ξ -allowable together with a fixed witness to

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P},$$

then

$$W[G] \models \text{“}\mathbb{Q}/G \Vdash (x, y) \in A_m\text{”}.$$

Proof. Let β be the stage at which the occurrence (x, y, m, ξ) is first entered into the auxiliary name. At that stage the construction is in clause (a), and by definition the selected real y is ξ -persistent for (x, m) over $\mathbb{P}_\beta$.

By Lemma 2.15, the final forcing $\mathbb{P}$ is ξ -allowable. Moreover

$$\mathbb{P} \succeq_{\text{res}} \mathbb{P}_\beta.$$

Applying the definition of ξ -persistence to this complete reservoir extension gives

$$W[G] \models (x, y) \in A_m.$$

If $\mathbb{Q}$ is as in the second assertion, then transitivity of $\succeq_{\text{res}}$ gives

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P}_\beta.$$

The same persistence statement at stage β therefore yields

$$W[G] \models \text{“}\mathbb{Q}/G \Vdash (x, y) \in A_m''\text{”},$$

which is the desired conclusion. $\square$

Lemma 2.18. *For every ordinal α , the class $\mathcal{A}_\alpha^W$ is nonempty.*

Proof. We argue by induction on α . The class $\mathcal{A}_0^W$ is nonempty. Suppose that $0 < \alpha$ and that ξ -allowability has been defined for every $\xi < \alpha$. Choose a bookkeeping function of positive countable length satisfying the displayed requirements, and use, for example, the canonical injective area assignment $\eta_\beta = \beta$, which belongs to L . The bookkeeping may be chosen to use check-names, so its reservoir support is empty and hence contained in U_β at every stage. At each stage, either the set $C_\beta^\alpha(K * G_\beta)$ is nonempty and clause (a) applies, or it is empty and clause (b) applies. In either case the next factor is an almost disjoint coding forcing of the form required in Definition 2.8. The recursion therefore produces an α -allowable iteration. $\square$

There are only set many isomorphism types of allowable forcings in W : every such forcing has size at most $\aleph_1$ and belongs to $H(\omega_2)^W$. Since the sequence

$$\langle \mathcal{A}_\alpha^W \mid \alpha \in \text{Ord} \rangle$$

is decreasing and each $\mathcal{A}_\alpha^W$ is nonempty, it eventually stabilizes. Let

$$\alpha_0 := \min \{ \alpha : \forall \beta \geq \alpha \ (\mathcal{A}_\beta^W = \mathcal{A}_\alpha^W) \}.$$

Set

$$\alpha^* := \alpha_0 + 1.$$

Definition 2.19. *An ∞ -allowable iteration is an α^* -allowable iteration. A forcing is ∞ -allowable if it is isomorphic to the final forcing of such an iteration.*

Since

$$\mathcal{A}_{\alpha^*}^W = \mathcal{A}_{\alpha_0}^W,$$

the class of ∞ -allowable forcings is the fixed class $\mathcal{A}_{\alpha_0}^W$. The use of the successor index α^* ensures that the stage rule also tests persistence under

the fixed class itself. Equivalently, in clause (a) of an ∞ -allowable iteration, we minimize over

$$\xi < \alpha^*.$$

Thus every tentative value entered into the auxiliary set of an ∞ -allowable iteration has rank

$$\xi < \alpha^*.$$

3 Definition of the universe in which the Π_3^1 uniformization property holds

The notion of ∞ -allowability will now be used to define the universe in which the Π_3^1 -uniformization property holds. Let W be our ground model. We construct an ω_1 -length iteration whose countable initial segments are ∞ -allowable over W .

Fix a bookkeeping function

$$F : \omega_1 \longrightarrow H(\omega_1)^W$$

such that

- F is surjective;
- for every $a \in H(\omega_1)^W$, the set $F^{-1}(\{a\})$ is unbounded in ω_1 .

Fix the canonical area assignment

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \omega_1 \rangle = \langle \beta \mid \beta < \omega_1 \rangle \in L.$$

Thus

$$U_\beta = \text{ran}(\vec{\eta} \restriction \beta) = \beta,$$

and the coordinate $\eta_\beta = \beta$ supplies the coding area used at stage β .

We define in W , by recursion on $\beta < \omega_1$, an iteration $\mathbb{P}_{\omega_1}$ together with a sequence of auxiliary names

$$\langle \dot{I}_\beta \mid \beta \leq \omega_1 \rangle.$$

No generic filter is chosen during this recursion. The notation below involving a generic $K * G_\beta$ is only a semantic description of the canonical names selected in W . At a limit stage $\lambda \leq \omega_1$, we keep the fixed reservoir $\mathbb{C}^{\text{area}}$ and take the finite-support limit of the almost disjoint iteration. We let $\dot{I}_\lambda$ be the canonical name forced to be the union of the earlier interpreted auxiliary names.

Suppose that $\beta < \omega_1$ and that $(\mathbb{P}_\beta, \dot{I}_\beta)$ has already been constructed. Assume that the value $F(\beta)$ decodes as a pair $(F(\beta)_0, F(\beta)_1)$ and that

$$F(\beta)_0 = (\eta_1, \eta_2, m), \quad \eta_1 \leq \beta, \quad m \in \omega.$$

Let $(\dot{x}, \dot{y})$ be the η_2 -th pair, in the canonical $<_L$ -order fixed above, of nice $\mathbb{P}_{\eta_1}$ -names in stage-respecting normal form; equivalently, their canonical reservoir support is contained in

$$U_{\eta_1} = \eta_1.$$

Lift these names canonically to $\mathbb{P}_\beta$. By Easton's lemma, every pair of reals in the $\mathbb{P}_{\eta_1}$ -extension has stage-respecting names in this normal form.

For the semantic description, let $K * G_\beta$ be $\mathbb{P}_\beta$ -generic over W . Let

$$\mathfrak{a}_\beta = (E_\beta, A_\beta) := \text{supp}_{\text{AP}}((\dot{x}, \dot{y}, \dot{m}), \mathbb{P}_\beta),$$

and put

$$G_{\mathfrak{a}_\beta} := K_{E_\beta} * G_{A_\beta}, \quad M_{\mathfrak{a}_\beta} := W[K_{E_\beta}][G_{A_\beta}].$$

Set

$$x = \dot{x}^{G_{\mathfrak{a}_\beta}}, \quad y = \dot{y}^{G_{\mathfrak{a}_\beta}}.$$

Assume that

$$W[K][G_\beta] \models (x, y) \in A_m.$$

We distinguish two cases.

- (1) *The first uniformization case.* There is a least ordinal $\zeta < \alpha^*$ for which some real

$$y_0 \in M_{\mathfrak{a}_\beta} \cap 2^\omega$$

satisfies

$$W[K][G_\beta] \models (x, y_0) \in A_m,$$

and no ζ -allowable complete reservoir extension of $\mathbb{P}_\beta$ forces (x, y_0) out of A_m . Among all such values, choose y_0 so that

$$\text{key}_{\mathfrak{a}_\beta}(\langle x, y_0, m \rangle)$$

is $<_L$ -least. Let $\dot{y}_0$ be the $<_L$ -least stage-respecting $\mathbb{P}_{\mathfrak{a}_\beta}$ -name for y_0 , lifted to $\mathbb{P}_\beta$. This name is used only to form the auxiliary name; the choice of y_0 was made by the semantic key.

In this case y_0 is declared to be a tentative value for $f_m(x)$ of rank ζ . If $y \neq y_0$, let

$$z_\beta^* := y.$$

If $y = y_0$, let z_β^* be the real $w \neq y_0$ in $M_{\mathfrak{a}_\beta}$ for which

$$\text{key}_{\mathfrak{a}_\beta}(\langle x, w, m \rangle)$$

is $<_L$ -least. Let $\dot{z}_\beta^*$ be the $<_L$ -least stage-respecting $\mathbb{P}_{\mathfrak{a}_\beta}$ -name for z_β^* , lifted to $\mathbb{P}_\beta$.

We let

$$\mathbb{P}(\beta) := \mathbb{A}_D(\dot{Y}(\dot{x}, \dot{z}_\beta^*, \dot{m}, \dot{g}_{\eta_\beta}))$$

be the corresponding coding forcing over the fixed reservoir and put

$$\mathbb{P}_{\beta+1} = \mathbb{P}_\beta * \mathbb{P}(\beta).$$

The auxiliary name $\dot{I}_{\beta+1}$ is the canonical name forced to satisfy

$$\dot{I}_{\beta+1}^{\dot{G}_{\beta+1}} = \dot{I}_\beta^{\dot{G}_\beta} \cup \{(\dot{x}^{\dot{G}_\beta}, \dot{y}_0^{\dot{G}_\beta}, \dot{m}, \check{\zeta})\}.$$

Thus the tentative value y_0 is not coded, while (x, z_β^*, m) is coded.

- (2) *The second uniformization case.* Suppose that no real in $M_{\mathfrak{a}_\beta} \cap 2^\omega$ passes the test at any rank $\zeta < \alpha^*$. In particular, the current real y can be forced out of A_m by an α^* -allowable complete reservoir continuation of the current iteration. We choose the $<_L$ -least such continuation, restrict its quotient below a condition forcing

$$(x, y) \notin A_m,$$

and append this quotient to the current iteration. The auxiliary name is continued by the auxiliary name supplied by this α^* -allowable continuation. No new tentative value is selected at this stage.

This defines a pair $(\mathbb{P}_{\omega_1}, \dot{I}_{\omega_1}) \in W$. The bookkeeping can be recoded into a function $F' \in W$ such that, for every $\delta < \omega_1$, the initial pair $(\mathbb{P}_\delta, \dot{I}_\delta)$ is α^* -allowable over W relative to

$$(F' \restriction \delta, \vec{\eta} \restriction \delta).$$

Here $\vec{\eta} \restriction \delta \in L$, and every bookkeeping name used before δ has reservoir support contained in the corresponding $U_\beta = \beta$.

Fact 3.1. *The pair $(\mathbb{P}_{\omega_1}, \dot{I}_{\omega_1})$ belongs to W , and every countable initial segment is α^* -allowable over W relative to a fixed bookkeeping function $F' \in W$ and the corresponding initial segment of the canonical area assignment $\vec{\eta} \in L$.*

Let G_{ω_1} be $\mathbb{P}_{\omega_1}$ -generic over W . For every $\beta \leq \omega_1$, let $G_\beta = G_{\omega_1} \cap \mathbb{P}_\beta$ and put

$$I_\beta := \dot{I}_\beta^{G_\beta}, \quad I := I_{\omega_1}.$$

Lemma 3.2. *Let $\beta \leq \gamma < \omega_1$, and suppose that $\mathbb{P}_\gamma$ is an initial segment of the final iteration extending $\mathbb{P}_\beta$. Let G_γ be $\mathbb{P}_\gamma$ -generic over W , and let*

$$x, y \in W[G_\beta] \cap 2^\omega, \quad m < \omega, \quad \xi \in \text{Ord}.$$

If y is not ξ -persistent for (x, m) over $\mathbb{P}_\beta$, then y is not ξ -persistent for (x, m) over $\mathbb{P}_\gamma$. Equivalently, if y is ξ -persistent for (x, m) over $\mathbb{P}_\gamma$, then y was already ξ -persistent for (x, m) over $\mathbb{P}_\beta$.

Proof. Suppose that y is not ξ -persistent over $\mathbb{P}_\beta$. After restricting below a condition, there is a ξ -allowable complete reservoir extension

$$\mathbb{Q} \succeq_{\text{res}} \mathbb{P}_\beta$$

such that, over the chosen copy of $\mathbb{P}_\beta$,

$$\mathbb{Q}/G_\beta \Vdash (x, y) \notin A_m.$$

Since $\mathbb{P}_\gamma$ is an initial segment of the final ∞ -allowable iteration, it is α^* -allowable. If $\xi < \alpha^*$, Lemma 2.15 implies that $\mathbb{P}_\gamma$ is ξ -allowable. If $\xi \geq \alpha^*$, then

$$\mathcal{A}_\xi^W = \mathcal{A}_{\alpha^*}^W$$

by the stabilization of the allowability hierarchy. Hence, in all cases, $\mathbb{P}_\gamma$ is ξ -allowable. There is a $\mathbb{P}_\beta$ -name $\dot{\mathbb{R}}_{\beta, \gamma}$ such that

$$\mathbb{P}_\gamma \cong \mathbb{P}_\beta * \dot{\mathbb{R}}_{\beta, \gamma}.$$

After identifying $\mathbb{P}_\beta$ with its chosen complete reservoir copy in $\mathbb{Q}$, there is also a $\mathbb{P}_\beta$ -name $\dot{\mathbb{T}}_Q$ such that

$$\mathbb{Q} \cong \mathbb{P}_\beta * \dot{\mathbb{T}}_Q,$$

and, whenever G_β is $\mathbb{P}_\beta$ -generic,

$$\dot{\mathbb{R}}_{\beta, \gamma}^{G_\beta} \cong \mathbb{P}_\gamma / G_\beta, \quad \dot{\mathbb{T}}_Q^{G_\beta} \cong \mathbb{Q} / G_\beta.$$

Over the common initial forcing form the fibered product

$$\mathbb{S}_0 := \mathbb{P}_\beta * (\dot{\mathbb{R}}_{\beta, \gamma} \times \dot{\mathbb{T}}_Q).$$

Apply the product construction from Lemma 2.16 relative to the common complete reservoir subiteration $\mathbb{P}_\beta$: after moving the fresh reservoir coordinates of the two tails to disjoint blocks, the amalgam has a normalized ξ -allowable presentation. Thus there are a normalized ξ -allowable iteration $\mathbb{S}$ and an isomorphism

$$\pi : \mathbb{S}_0 \cong \mathbb{S}.$$

Under π , the canonical copy of $\mathbb{P}_\gamma$ in $\mathbb{S}_0$ becomes a complete reservoir subiteration of $\mathbb{S}$. The quotient of $\mathbb{S}$ over this copy contains the translated copy of $\dot{\mathbb{T}}_Q$. Since

$$\dot{\mathbb{T}}_Q^{G_\beta} \Vdash (x, y) \notin A_m$$

and $(x, y) \notin A_m$ is a Σ_3^1 statement, the same witness remains valid after adjoining the generic for $\dot{\mathbb{R}}_{\beta, \gamma}^{G_\beta}$, by Shoenfield absoluteness applied to the Π_2^1 matrix. Consequently,

$$\mathbb{S}/G_\gamma \Vdash (x, y) \notin A_m.$$

Thus $\mathbb{S}$ is a ξ -allowable complete reservoir extension of $\mathbb{P}_\gamma$ witnessing that y is not ξ -persistent over $\mathbb{P}_\gamma$, as required. $\square$

By Lemma 2.17, every tentative value for $f_m(x)$ of rank below α^* belongs to A_m in the final model. It remains to show that, for every $m \in \omega$ and every real x with $A_{m,x} \neq \emptyset$, there is exactly one y such that

$$(x, y) \in A_m \quad \text{and} \quad (x, y, m) \text{ is not coded into } \vec{S}.$$

The next lemma gives the required dichotomy.

Lemma 3.3. *In $W[G_{\omega_1}]$ the following dichotomy holds true:*

1. *The auxiliary set contains a tentative value for (x, m) , that is,*

$$\exists y \in 2^\omega \exists \xi < \alpha^* ((x, y, m, \xi) \in I).$$

In this case there is a unique real y_0 such that

$$W[G_{\omega_1}] \models (x, y_0) \in A_m \wedge (x, y_0, m) \text{ is not coded into } \vec{S}.$$

2. *The auxiliary set contains no tentative value for (x, m) , that is,*

$$\forall y \in 2^\omega \forall \xi < \alpha^* ((x, y, m, \xi) \notin I).$$

In this case

$$W[G_{\omega_1}] \models A_{m,x} = \emptyset.$$

Proof. Assume first that the auxiliary set contains a tentative value for (x, m) . Let $\xi_0 < \alpha^*$ be the least rank for which some stage selects a real y for (x, m) and enters (x, y, m, ξ_0) into the auxiliary set. Among all such stages of rank ξ_0 , choose one for which the key used at that stage is $<_L$ -least. Let y_0 be the value selected there, let κ_0 be this key, and let β be the least stage at which y_0 is selected with rank ξ_0 and key κ_0 .

Claim. $W[G_{\omega_1}] \models (x, y_0) \in A_m$.

Proof of the first claim. This follows immediately from Lemma 2.17. $\square$

Claim (No reversal). *Let η be a stage at which the local datum is again (x, m) and whose local admissible support is $\mathfrak{a}_\eta$. Suppose that $y_0 \in M_{\mathfrak{a}_\eta}$ and that y_0 is ξ_0 -persistent for (x, m) over $\mathbb{P}_\eta$. Then the construction cannot select a value $v \neq y_0$ for (x, m) at stage η .*

Proof of the no-reversal claim. Since y_0 passes the persistence test at rank ξ_0 , the second uniformization case does not apply at stage η . Let v be the value selected in the first uniformization case and let ρ be its rank. Then $\rho \leq \xi_0$.

If $\rho < \xi_0$, then (x, v, m, ρ) is entered into the auxiliary set, contradicting the minimality of ξ_0 . Hence $\rho = \xi_0$. Assume that $v \neq y_0$. At stage η , the real

y_0 is one of the rank- ξ_0 reals compared by key. Therefore the selection of v gives

$$\text{key}_{\mathfrak{a}_\eta}(\langle x, v, m \rangle) <_L \text{key}_{\mathfrak{a}_\eta}(\langle x, y_0, m \rangle).$$

Passing to a common admissible closure of the support used at stage η and the support used when y_0 was selected, Lemma 2.13 shows that

$$\text{key}_{\mathfrak{a}_\eta}(\langle x, y_0, m \rangle) = \kappa_0.$$

The same support-independence identifies the key of v at stage η with the key carried by the occurrence (x, v, m, ξ_0) in the auxiliary set. Thus there is an occurrence of rank ξ_0 with key $<_L \kappa_0$, contradicting the choice of κ_0 . Hence $v = y_0$. $\square$

Claim.

$W[G_{\omega_1}] \models \text{"}y_0 \text{ is the unique real } y \text{ such that}$
 $(x, y, m) \text{ is not coded somewhere in the } \vec{S}\text{-sequence.}"$

Proof of the second claim. We first show that (x, y_0, m) is not coded into $\vec{S}$. Suppose towards a contradiction that stage η codes (x, y_0, m) . Then y_0 belongs to the local model used at that stage.

If $\eta < \beta$, then y_0 is ξ_0 -persistent over $\mathbb{P}_\eta$. Indeed, otherwise Lemma 3.2 would imply that y_0 is not ξ_0 -persistent over $\mathbb{P}_\beta$, contradicting its selection at stage β . If $\eta \geq \beta$, then (x, y_0, m, ξ_0) is already in the auxiliary set of the initial segment $\mathbb{P}_\eta$, and Lemma 2.17 gives that y_0 is ξ_0 -persistent over $\mathbb{P}_\eta$.

Thus the no-reversal claim applies at stage η . The stage cannot be in the second uniformization case, and in the first uniformization case it must select y_0 . But a stage which selects y_0 does not code (x, y_0, m) : if the bookkeeping value is y_0 , the definition uses the alternate value $z_\eta^* \neq y_0$, and otherwise it codes the bookkeeping value. This is a contradiction. Hence (x, y_0, m) is not coded into $\vec{S}$.

We now prove uniqueness. Let $y \neq y_0$ and suppose that

$$W[G_{\omega_1}] \models (x, y) \in A_m.$$

Choose a stage $\eta \geq \beta$ at which the bookkeeping presents stage-respecting names for x , y , and m , and such that the local admissible support also contains a support for y_0 . By the downward form of Shoenfield absoluteness applied to the Π_3^1 definition of A_m , the model $W[G_\eta]$ already satisfies $(x, y) \in A_m$. Since (x, y_0, m, ξ_0) belongs to the auxiliary set of the initial segment $\mathbb{P}_\eta$, Lemma 2.17 shows that y_0 is ξ_0 -persistent over $\mathbb{P}_\eta$. The no-reversal claim therefore implies that the stage selects y_0 . Since the bookkeeping value at this stage is $y \neq y_0$, the construction codes (x, y, m) into $\vec{S}$.

This proves that y_0 is the unique real y such that $(x, y) \in A_m$ and (x, y, m) is not coded into $\vec{S}$, and finishes the proof in the first case of the dichotomy. $\square$

We now assume that the auxiliary set contains no tentative value for (x, m) . We must show that $W[G_{\omega_1}] \models$ “The x -section of A_m is empty”.

Under this assumption, whenever we are at a stage β such that there is a y such that $F(\beta) = (x, y, m)$, then the second uniformization case in the definition of $\mathbb{P}_{\omega_1}$ must apply. For every such y , the stage β construction appends an ∞ -allowable eliminating tail which forces

$$(x, y) \notin A_m.$$

By Lemma 3.2, this eliminability is stable under all later ∞ -allowable continuations. Since $(x, y) \notin A_m$ is Σ_3^1 , upward absoluteness, equivalently Shoenfield absoluteness for its Π_2^1 matrix, gives

$$W[G_{\omega_1}] \models \neg \exists y ((x, y) \in A_m).$$

This finishes the proof of the lemma. □

Corollary 3.4. *In $W[G_{\omega_1}]$, the Π_3^1 -uniformization property holds. More precisely, for every $m \in \omega$, define a relation U_m on $2^\omega \times 2^\omega$ by*

$$U_m(x, y) \iff (x, y) \in A_m \wedge \neg \Phi(\langle x, y, m \rangle).$$

Then U_m is the graph of a Π_3^1 uniformizing function for A_m .

Proof. The relation $(x, y) \in A_m$ is Π_3^1 , while $\Phi(\langle x, y, m \rangle)$ is Σ_3^1 . Hence U_m is Π_3^1 . By the preceding dichotomy, whenever the section $A_{m,x}$ is nonempty, there is exactly one $y \in A_{m,x}$ for which $\Phi(\langle x, y, m \rangle)$ fails; if the section is empty, there is no such y . Therefore U_m is the graph of a function with domain $\text{pr}_1(A_m)$, and this graph is contained in A_m . □

4 Forcing over the canonical inner models M_n

Fix for the remainder of this section a natural number n with $1 \leq n < \omega$, and assume that $M_n^\#$ exists. As always we let M_n be the resulting inner model one obtains when iterating the sharp out of the universe. Our goal is to prove the following theorem.

Theorem. *There is a set-generic extension of M_n in which the Π_{n+3}^1 -uniformization property holds.*

The proof follows the construction over L from the preceding sections, but three ingredients require M_n -analogues: the recovery of the canonical lower part $M_n|_{\omega_1}$, the definition of the coding predicate, and the absoluteness argument which keeps tentative uniformizing values in their projective sections. We begin with the inner-model-theoretic background needed for these analogues.

4.1 Inner-model-theoretic background and generic absoluteness

We use the notation and comparison conventions of Steel's analysis of mice with finitely many Woodin cardinals; see [21, 22]. An *ordinary premouse* means a premouse in the parameter-free premouse language. In particular, no real or other base set is included as a predicate. Every proper initial segment of M_n is n -small and ω -sound, and the background construction of M_n supplies the realizability strategy used on such an initial segment in a comparison.

Definition 4.1. *Let $\mathcal{M}$ be a premouse and let $\eta < \text{Ord}^{\mathcal{M}}$. We say that $\mathcal{M}$ is n -small above η if, whenever E is an extender on the $\mathcal{M}$ -sequence and*

$$\eta < \text{crit}(E),$$

the initial segment $\mathcal{J}_{\text{crit}(E)}^{\mathcal{M}}$ does not have n Woodin cardinals above η . We say that $\mathcal{M}$ is n -small if it is n -small above 0.

The weak iterability notion used at the finite projective levels is Steel's Π_n -iterability. Its recursive definition depends on the parity of n . At even levels one uses the corresponding finite weak iteration game, while at odd levels the last round includes the appropriate goodness requirement for the branch selected by player II. We use the standard game from [21, 22] and isolate the notation needed below.

Definition 4.2. *Let $\mathcal{M}$ be a countable premouse and let $\eta < \text{Ord}^{\mathcal{M}}$ be a cutpoint. We say that $\mathcal{M}$ is Π_n -iterable above η if player II has a winning strategy in Steel's game*

$$\mathcal{G}(\mathcal{M}, \eta, n)$$

for Π_n -iterability above η . We say simply that $\mathcal{M}$ is Π_n -iterable if it is Π_n -iterable above 0.

Steel's analysis gives the following complexity statement.

Fact 4.3. *For each fixed $1 \leq n < \omega$, the relation*

$$“x \text{ codes a countable premouse which is } \Pi_n\text{-iterable}”$$

is Π_{n+1}^1 in the real code x . Moreover, every countable realizable n -small premouse is Π_n -iterable.

We shall use Π_n -iterability only through the following comparison consequence.

Lemma 4.4. *Let $\mathcal{M}$ and $\mathcal{N}$ be countable ordinary premice. Assume that both are ω -sound and satisfy*

$$\rho_\omega(\mathcal{M}) = \rho_\omega(\mathcal{N}) = \omega,$$

that $\mathcal{M} \triangleleft M_n$, and that $\mathcal{N}$ is n -small and Π_n -iterable. Then the comparison of $\mathcal{M}$ with $\mathcal{N}$ is successful. Consequently,

$$\mathcal{M} \leq \mathcal{N} \quad \text{or} \quad \mathcal{N} \leq \mathcal{M}.$$

Proof. This is the parameter-free, lower-part case of Steel's comparison theorem for n -small mice. Run the usual least-disagreement coiteration. The $\mathcal{M}$ -side is governed by the realization strategy inherited from the background construction of M_n , while the $\mathcal{N}$ -side is governed by the strategy witnessing Π_n -iterability. Steel's branch uniqueness and $\mathcal{Q}$ -structure analysis show that the required branches at limit stages are well-founded and that the coiteration terminates. Since both premices are ω -sound and project to ω , the standard no-drop pullback argument gives the displayed initial-segment relation between the original premices. See [21, 22]. $\square$

We next specify the generic extensions in which the comparison lemma will be used. All cardinalities in the following definition are computed in M_n .

Definition 4.5. *A model N is a small M_n -generic extension if*

$$N = M_n[G]$$

for some cardinal-preserving forcing $\mathbb{P} \in M_n$ such that $M_n \models |\mathbb{P}| \leq \aleph_1$ and some M_n -generic filter $G \subseteq \mathbb{P}$. A small two-step extension is an extension

$$M_n[G] \subseteq M_n[G][H]$$

*obtained from a two-step forcing $\mathbb{P} * \dot{\mathbb{Q}} \in M_n$ which is forced to be cardinal preserving and of size at most $\aleph_1$.*

The preparatory branch forcing and every forcing iteration used later will be shown to have size $\aleph_1$ and to preserve cardinals. Hence all models which occur in the proof are small M_n -generic extensions in the sense of Definition 4.5.

Lemma 4.6. *Let N be a small M_n -generic extension. If $\mathcal{M} \triangleleft M_n$ is countable in N , then*

$$N \models \text{"}\mathcal{M} \text{ is } \Pi_n\text{-iterable."}$$

Proof. The background construction of M_n realizes every proper initial segment of M_n . The standard preservation argument for Steel's weak iteration games shows that this realization continues to supply the required branches in the small, ω_1 -preserving generic extensions considered here. Thus the Π_n -iterability of a genuine initial segment of M_n is preserved. We will use only this preservation statement; we do not claim that the class of all real codes for Π_n -iterable premices is absolute between arbitrary outer models with the same ω_1 . $\square$

The second background ingredient is the mouse-operator closure which yields projective generic absoluteness. Recall that a set is *self-wellordered*, abbreviated *swo*, if it codes a wellorder of itself. For a self-wellordered set X , let $M_k^\#(X)$ denote the canonical active X -mouse with k Woodin cardinals. Thus $M_0^\#(X)$ is the usual sharp over X .

Fact 4.7. *The model M_n is closed under the operator*

$$X \mapsto M_{n-1}^\#(X)$$

below its least Woodin cardinal. In particular, for every self-wellordered $X \in H(\omega_2)^{M_n}$, the mouse $M_{n-1}^\#(X)$ exists and is normally ω_2 -iterable in M_n .

This closure is preserved by the small forcing notions which occur in the construction.

Lemma 4.8. *Let $N = M_n[G]$ be a small M_n -generic extension. Then*

$$N \models \text{“} M_{n-1}^\#(X) \text{ exists and is normally } \omega_2\text{-iterable} \\ \text{for every swo } X \in H(\omega_2)\text{.”}$$

Moreover, whenever X belongs to both M_n and N , the two models compute the same mouse $M_{n-1}^\#(X)$.

Proof. Apply the generic preservation theorem for mouse operators to $M_{n-1}^\#$. In the notation of [20, Lemma 3.4], take $\kappa = \omega_1$ and replace the index occurring there by $n - 1$. Fact 4.7 supplies the hypothesis that $M_{n-1}^\#(X)$ exists for every self-wellordered $X \in H(\omega_2)^{M_n}$. The cited lemma gives existence, normal ω_2 -iterability, and correctness of the operator in the forcing extension. $\square$

We can now state the precise generic absoluteness principle used later. It is the two-step form which permits real parameters introduced by the first forcing.

Theorem 4.9. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension. Then

$$M_n[G] \prec_{\Sigma_{n+2}^1} M_n[G][H].$$

Equivalently, for every real parameter $a \in M_n[G]$ and every Σ_{n+2}^1 formula $\vartheta(v)$,

$$M_n[G] \models \vartheta(a) \iff M_n[G][H] \models \vartheta(a).$$

The same equivalence holds for Π_{n+2}^1 formulas.

Proof. By Lemma 4.8, the intermediate model $M_n[G]$ is closed under $X \mapsto M_{n-1}^\#(X)$ for self-wellordered $X \in H(\omega_2)^{M_n[G]}$. Apply the Martin–Solovay–Schindler generic absoluteness theorem in the form of [20, Lemma 3.7], with the mouse index $n - 1$ and $\kappa = \omega_1$, to the second forcing. This yields Σ_{n+2}^1 -elementarity between $M_n[G]$ and $M_n[G][H]$. The Π_{n+2}^1 assertion follows by taking complements. $\square$

The point at which Theorem 4.9 enters the uniformization proof is recorded in the following consequence.

Corollary 4.10. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension, let $a \in M_n[G]$ be a real, and let $\varphi(v)$ be a Σ_{n+3}^1 formula. If

$$M_n[G] \models \varphi(a),$$

then

$$M_n[G][H] \models \varphi(a).$$

Consequently, if $A \subseteq (2^\omega)^2$ is Π_{n+3}^1 and

$$M_n[G] \models (x, y) \notin A,$$

then $(x, y) \notin A$ remains true in $M_n[G][H]$.

Proof. Write

$$\varphi(a) \equiv \exists z \psi(z, a),$$

where ψ is Π_{n+2}^1 . Choose a witness $z \in M_n[G]$. By Theorem 4.9, the statement $\psi(z, a)$ remains true in $M_n[G][H]$, so the same witness establishes $\varphi(a)$ there. The final assertion is the special case in which $\varphi(x, y)$ defines the complement of A . $\square$

4.2 The definable lower part of M_n

For the coding construction we do not need to recover the whole proper class model M_n . It is enough to recover the canonical initial segment $M_n|_{\omega_1}$ uniformly in the generic extensions which occur in the proof. To this end we isolate a projectively definable cofinal system of countable lower-part initial segments of M_n .

The lower-part requirement in the next definition is important. At the higher projective levels there can be nonstandard Π_n -iterable premice which do not occur on the M_n -sequence. Since all ordinals below $\omega_1^{M_n}$ lie below the least Woodin cardinal of M_n , the genuine initial segments relevant to the present coding argument have no Woodin cardinal. Restricting to such premice removes the unwanted objects.

Definition 4.11. Let $N = M_n[G]$ be a small M_n -generic extension in the sense of Definition 4.5. In N , let $\mathcal{I}_n^N$ be the class of all countable premice $\mathcal{M}$ such that

1. $\mathcal{M}$ is a passive ordinary premouse;
2. $\mathcal{M}$ is a lower-part premouse, that is, $\mathcal{M}$ has no Woodin cardinal;
3. $\mathcal{M}$ is n -small and ω -sound, and

$$\rho_\omega(\mathcal{M}) = \omega;$$

4. $\mathcal{M}$ is Π_n -iterable in the sense of Definition 4.2.

We identify a real coding such a premouse with its transitive collapse. When there is no danger of confusion about the ambient extension, we simply write $\mathcal{I}_n$.

Lemma 4.12. Let $N = M_n[G]$ be a small M_n -generic extension. Then the set of real codes for members of $\mathcal{I}_n^N$ is Π_{n+1}^1 -definable, uniformly in N . Moreover, every member of $\mathcal{I}_n^N$ is of the form

$$\mathcal{J}_\eta^{M_n}$$

for some $\eta < \omega_1$, and the set

$$C_n := \{\eta < \omega_1 : \mathcal{J}_\eta^{M_n} \in \mathcal{I}_n^N\}$$

is cofinal in ω_1 .

Proof. The assertion that a real codes a well-founded countable passive ordinary premouse has complexity at most Π_1^1 . Once the premouse has been decoded, the requirements that it be lower-part, n -small, ω -sound, and have $\rho_\omega = \omega$ are given by the usual fine-structural conditions and do not increase the complexity beyond Π_1^1 . By Fact 4.3, the remaining iterability requirement is Π_{n+1}^1 . Since $n \geq 1$, the conjunction is Π_{n+1}^1 . The same formula is used in every small M_n -generic extension, which gives uniformity.

We next show that no nonstandard lower-part premouse belongs to $\mathcal{I}_n^N$. Let $\mathcal{M} \in \mathcal{I}_n^N$. Since $\mathcal{M}$ is countable in N and the forcing from M_n to N preserves ω_1 , we have

$$\text{Ord}^{\mathcal{M}} < \omega_1^{M_n}.$$

Choose an ordinal η such that

$$\text{Ord}^{\mathcal{M}} < \eta < \omega_1$$

and such that $\mathcal{J}_\eta^{M_n}$ is passive, lower-part, ω -sound, and projects to ω . Standard fine structure gives cofinally many such ordinals $\eta < \omega_1$. The premouse

$\mathcal{J}_\eta^{M_n}$ is realizable in the background construction of M_n , and it remains Π_n -iterable in N by Lemma 4.6.

Apply Lemma 4.4 to $\mathcal{J}_\eta^{M_n}$ and $\mathcal{M}$. We obtain

$$\mathcal{J}_\eta^{M_n} \trianglelefteq \mathcal{M} \quad \text{or} \quad \mathcal{M} \trianglelefteq \mathcal{J}_\eta^{M_n}.$$

The first alternative is impossible because $\text{Ord}^{\mathcal{M}} < \eta$. Hence

$$\mathcal{M} \trianglelefteq \mathcal{J}_\eta^{M_n},$$

so $\mathcal{M}$ is itself an initial segment of M_n . Thus $\mathcal{M} = \mathcal{J}_\xi^{M_n}$ for some $\xi < \omega_1$.

Conversely, suppose that $\eta < \omega_1$ and that $\mathcal{J}_\eta^{M_n}$ is passive, lower-part, ω -sound, and projects to ω . It is n -small and realizable in the construction of M_n . By Lemma 4.6, it is Π_n -iterable in N , and hence

$$\mathcal{J}_\eta^{M_n} \in \mathcal{I}_n^N.$$

Since the ordinals η with these fine-structural properties are cofinal in ω_1 , the set C_n is cofinal in ω_1 . $\square$

We can now recover $M_n|_{\omega_1}$ without referring to the proper class M_n in the defining formula. In a small generic extension N , let $\Theta_n(X)$ be the following assertion:

$$\begin{aligned} \Theta_n(X) \iff & X \text{ is transitive and } \text{Ord} \cap X = \omega_1, \\ & \text{and for every } z, \\ & z \in X \iff \exists \mathcal{M} (\mathcal{M} \in \mathcal{I}_n^N \wedge z \in \mathcal{M}). \end{aligned}$$

Here the quantifier over $\mathcal{M}$ may equivalently be read as a quantifier over real codes for countable premice satisfying Definition 4.11.

Definition 4.13. *Let $N = M_n[G]$ be a small M_n -generic extension. We say in N that*

$$X = M_n|_{\omega_1}$$

if $\Theta_n(X)$ holds. Equivalently,

$$X = \bigcup \mathcal{I}_n^N.$$

Lemma 4.14. *Let $N = M_n[G]$ be a small M_n -generic extension. Then there is a unique X such that $N \models \Theta_n(X)$, and this set is the true initial segment*

$$X = M_n|_{\omega_1} = \mathcal{J}_{\omega_1}^{M_n}.$$

Consequently, the same formula Θ_n recovers the same set $M_n|_{\omega_1}$ in every small M_n -generic extension.

Proof. By Lemma 4.12, every member of $\mathcal{I}_n^N$ is an initial segment $\mathcal{J}_\eta^{M_n}$ for some $\eta < \omega_1$. Therefore

$$\bigcup \mathcal{I}_n^N \subseteq \mathcal{J}_{\omega_1}^{M_n}.$$

The same lemma states that the set of $\eta < \omega_1$ for which $\mathcal{J}_\eta^{M_n} \in \mathcal{I}_n^N$ is cofinal in ω_1 . Hence

$$\mathcal{J}_{\omega_1}^{M_n} \subseteq \bigcup \mathcal{I}_n^N.$$

Thus

$$\bigcup \mathcal{I}_n^N = \mathcal{J}_{\omega_1}^{M_n}.$$

This union is transitive and has ordinal height ω_1 , so it satisfies Θ_n . Conversely, the last clause of Θ_n uniquely determines X to be this union. Since the argument applies in every small generic extension, the recovery is uniform. $\square$

Corollary 4.15. *In every small M_n -generic extension and for every real x ,*

$$x \in M_n|_{\omega_1} \iff \exists \mathcal{M} \in \mathcal{I}_n (x \in \mathcal{M}).$$

In particular, the set of reals belonging to $M_n|_{\omega_1}$ has a uniform Σ_{n+2}^1 definition in these extensions.

Proof. The equivalence follows immediately from Lemma 4.14. In terms of real codes, the right-hand side begins with an existential real quantifier and then asserts a Π_{n+1}^1 property of the code, together with the arithmetic assertion that the decoded premouse contains x . It is therefore Σ_{n+2}^1 . $\square$

Local use of the recovery formula. The uniformity asserted above concerns the actual outer models of M_n used in the construction. It does not say that an arbitrary countable transitive model which internally evaluates the formula Θ_n must recover the true $M_n|_{\omega_1}$. Nor do we claim that the collection of real codes belonging to $\mathcal{I}_n$ is literally the same in every outer model. New reals may provide new codes and new putative iteration trees. What is absolute for the extensions under consideration is the union of the premisses which pass the test: it is always the true $\mathcal{J}_{\omega_1}^{M_n}$.

For this reason, the localized coding predicate introduced below will not ask a countable auxiliary model to reconstruct the global union on its own. Instead, it will be supplied with a countable premouse m which, from the outside, belongs to $\mathcal{I}_n$ and satisfies

$$\omega_1^m = \omega_1^M,$$

where M is the auxiliary countable transitive model. The premouse m then provides the correct local approximation to all canonical objects constructed from $M_n|_{\omega_1}$.

4.3 Canonical parameters from $M_n|_{\omega_1}$

We next isolate the canonical objects from the lower part of M_n which will be used by the coding construction. They are the M_n -analogue of the canonical constructible parameters used in Sections 2 and 3: a diamond sequence, an independent sequence of Suslin trees, an almost disjoint family, and a map which turns proper initial segments of an M_n -Cohen subset of ω_1 into coding areas. The point of the present subsection is not only that these objects exist in M_n , but that they are defined uniformly from the lower part recovered in Subsection 4.2. Consequently the same definitions are available in every small M_n -generic extension occurring below.

We first record the canonical order which will be used to make the choices in all the ensuing recursions.

Fact 4.16. *The order of construction of M_n induces a canonical well-order $<_{M_n}$ of M_n . Its restriction to the reals of M_n is a Δ^1_{n+2} well-order. Moreover, the restriction of this order to $M_n|_{\omega_1}$ is uniformly definable over the structure $M_n|_{\omega_1}$.*

Every real of M_n belongs to $M_n|_{\omega_1}$. Hence, whenever a subset of a countable ordinal or a countable structure is to be compared using $<_{M_n}$, we use the fixed canonical real code of that object and compare those codes. We shall use $<_{M_n}$ only to make canonical choices among objects already belonging to M_n ; it will never be used to choose a member of a projective section in the final generic extension.

The proof of the diamond principle uses the following form of condensation. We state precisely the consequence needed for the countable hulls appearing below.

Theorem 4.17. *Let $\mathcal{P} \trianglelefteq M_n$ be an ω -sound initial segment and let*

$$\pi : \bar{\mathcal{N}} \longrightarrow \mathcal{P}$$

be the inverse of the transitive collapse of a sufficiently elementary substructure of $\mathcal{P}$. Suppose that

$$\text{crit}(\pi) = \rho_\omega(\bar{\mathcal{N}}) = \omega_1^{\bar{\mathcal{N}}}.$$

Then one of the following alternatives holds.

1. $\bar{\mathcal{N}} \trianglelefteq M_n$.
2. *There are an initial segment $\mathcal{Q} \trianglelefteq M_n$ and an extender E on the M_n -sequence such that*

$$\text{lh}(E) = \omega_1^{\bar{\mathcal{N}}}$$

and $\bar{\mathcal{N}}$ is an initial segment of the degree-zero ultrapower $\text{Ult}_0(\mathcal{Q}, E)$.

For the hulls used in the proof of Lemma 4.18, the second alternative is impossible. Hence their transitive collapses are initial segments of M_n .

Proof. The dichotomy is the corresponding instance of Steel's condensation theorem for the fine-structural construction of M_n ; see [22]. We explain why the ultrapower alternative is excluded in the application below. The hulls are chosen so that the standard projectum of the collapse is its internal first uncountable cardinal

$$\lambda = \omega_1^{\mathcal{N}}.$$

The lower part $\mathcal{N} \upharpoonright \lambda$ therefore computes λ as its first uncountable cardinal. If the second alternative held, then an extender on the relevant M_n -sequence would be indexed at λ . In the corresponding extender model the index of such an extender is a cardinal with the closure properties supplied by the extender; in particular it cannot simultaneously be computed by the lower part as its first uncountable cardinal. This is the standard contradiction which rules out the ultrapower alternative in Jensen's diamond argument. Thus only the initial-segment alternative remains. $\square$

We can now carry out Jensen's canonical diamond construction inside M_n .

Lemma 4.18. *There is a sequence*

$$\vec{D}^n = \langle D_\alpha^n \mid \alpha < \omega_1 \rangle \in M_n$$

such that $D_\alpha^n \subseteq \alpha$ for every $\alpha < \omega_1$ and

$$M_n \models \text{“}\vec{D}^n \text{ is a } \diamond_{\omega_1}\text{-sequence.”}$$

The sequence is obtained by a recursion which is uniform over $M_n|_{\omega_1}$.

Proof. Work in M_n . We recursively define $\langle D_\beta^n \mid \beta < \alpha \rangle$ for $\alpha < \omega_1$. At successor stages put $D_\alpha^n = \emptyset$. Suppose that α is a nonzero limit ordinal and the sequence below α has already been defined. Call a pair (A, C) a *counterexample at α* if

$$A \subseteq \alpha, \quad C \subseteq \alpha \text{ is club in } \alpha,$$

and

$$\forall \beta \in C \ (D_\beta^n \neq A \cap \beta).$$

If such a pair exists in M_n , let (A_α, C_α) be the pair whose canonical real code is $<_{M_n}$ -least, and put

$$D_\alpha^n = A_\alpha.$$

If no such pair exists, put $D_\alpha^n = \emptyset$.

The recursion is uniform over $M_n|_{\omega_1}$. Indeed, every object considered at stage α is coded by a real of M_n , and all reals of M_n , together with the restriction of the construction order to those reals, belong to or are uniformly definable over $M_n|_{\omega_1}$.

Suppose toward a contradiction that $\vec{D}^n$ is not a diamond sequence in M_n . Let (A, C) be the $<_{M_n}$ -least counterexample, so that $A \subseteq \omega_1$, $C \subseteq \omega_1$ is club, and

$$\forall \alpha \in C \ (D_\alpha^n \neq A \cap \alpha).$$

Choose an ω -sound initial segment $\mathcal{J}_\theta^{M_n}$ containing A , C , $\vec{D}^n$, and the relevant part of the canonical construction order. Take a countable elementary substructure

$$X \prec \mathcal{J}_\theta^{M_n}$$

containing these parameters and such that

$$\alpha := X \cap \omega_1 \in C.$$

Let

$$\pi : \bar{X} \longrightarrow X$$

be the inverse of the transitive collapse. The hull may be chosen so that

$$\text{crit}(\pi) = \rho_\omega(\bar{X}) = \omega_1^{\bar{X}} = \alpha.$$

By Theorem 4.17, the collapse $\bar{X}$ is an initial segment of M_n .

The canonical construction order and the recursion defining $\vec{D}^n$ are therefore computed correctly in $\bar{X}$. The collapse fixes all ordinals below α and sends A and C to $A \cap \alpha$ and $C \cap \alpha$, respectively. By elementarity, $\bar{X}$ regards $(A \cap \alpha, C \cap \alpha)$ as the least counterexample to the sequence below α . Consequently the recursion at stage α gives

$$D_\alpha^n = A \cap \alpha.$$

This contradicts $\alpha \in C$. Hence $\vec{D}^n$ is a $\diamond_{\omega_1}$ -sequence. $\square$

We fix from now on the sequence supplied by Lemma 4.18. The phrase *the canonical M_n -diamond sequence* always refers to this fixed sequence.

Lemma 4.19. *There is a first-order formula $\text{CanDiamond}_n(X, M)$ such that, whenever $N = M_n[G]$ is a small M_n -generic extension and*

$$M = M_n|_{\omega_1}$$

is the set recovered in N by Definition 4.13, there is a unique X for which

$$N \models \text{CanDiamond}_n(X, M),$$

and this X is the true sequence $\vec{D}^n$.

Proof. The formula says that X is obtained by the recursion in the proof of Lemma 4.18, using the construction order on the reals of M . By Lemma 4.14, every small generic extension recovers the same transitive structure $M_n|_{\omega_1}$. Fact 4.16 then gives the same construction order, and transfinite recursion gives a unique sequence. Since the defining recursion is the one carried out in M_n , that sequence is $\vec{D}^n$. $\square$

The diamond sequence is used to construct the sequence of trees into which reals will later be written.

Lemma 4.20. *There is a sequence*

$$\vec{S}^n = \langle S_\xi^n \mid \xi < \omega_1 \rangle \in M_n$$

of normal Suslin trees such that, for every nonempty finite set $e \subseteq \omega_1$, the finite product

$$\prod_{\xi \in e} S_\xi^n$$

is a Suslin tree. The sequence is obtained by Jensen's simultaneous tree construction from $\vec{D}^n$, with every choice made $<_{M_n}$ -least. In particular, it is uniformly definable from $M_n|_{\omega_1}$ and $\vec{D}^n$.

Proof. This is Jensen's simultaneous Suslin-tree construction from the proof of Lemma 2.5, carried out inside M_n and using the canonical M_n -diamond sequence $\vec{D}^n$ from Lemma 4.18. At successor stages we preserve normality and splitting, and at limit stages the diamond guesses are used to seal maximal antichains in finite products. All choices are made $<_{M_n}$ -least.

The same argument as in the L -case shows that every finite product

$$\prod_{\xi \in e} S_\xi^n$$

is Suslin. Uniformity from $M_n|_{\omega_1}$ follows because $\vec{D}^n$ and the restriction of the construction order are uniformly recoverable from $M_n|_{\omega_1}$. $\square$

We identify every S_ξ^n with a subset of ω_1 using the canonical M_n -coding of countable normal trees. As in the constructible case, we shall say that a tree S_ξ^n is *destroyed* when a cofinal generic branch through it is added.

Corollary 4.21. *There is a first-order formula $\text{CanSuslin}_n(X, M)$ such that, in every small M_n -generic extension, the parameter $M = M_n|_{\omega_1}$ uniquely determines the true sequence $X = \vec{S}^n$.*

Proof. First recover $\vec{D}^n$ by Lemma 4.19, and then perform the canonical simultaneous tree recursion from Lemma 4.20. Both recursions are unique and use only the recovered lower part and its construction order. $\square$

We finish by fixing the two further canonical parameters needed for almost disjoint coding and for the coding areas. Canonical fine-structural mice satisfy GCH, and hence

$$M_n \models |({}^{<\omega_1}2)^{M_n}| = \omega_1.$$

Definition 4.22. The canonical M_n -almost disjoint family is

$$\mathcal{D}^n = \langle d_\xi^n \mid \xi < \omega_1 \rangle,$$

where, recursively, d_ξ^n is the $<_{M_n}$ -least infinite subset of ω which is almost disjoint from every d_ζ^n with $\zeta < \xi$.

The canonical M_n -area map is the $<_{M_n}$ -least bijection

$$\varrho_n : (<^{\omega_1} 2)^{M_n} \longrightarrow \omega_1.$$

Equivalently, after identifying countable binary sequences with countable subsets of ω_1 by the fixed coding, we may regard ϱ_n as a bijection

$$\varrho_n : ([\omega_1]^\omega)^{M_n} \longrightarrow \omega_1.$$

At stage $\xi < \omega_1$ of the first recursion there are only countably many previous members of $\mathcal{D}^n$, so the required infinite set exists. The second object exists by the displayed cardinal computation. Both are uniformly definable from $M_n|_{\omega_1}$ and the canonical construction order. We use the letter ϱ_n , rather than ρ , in order not to confuse the area map with the ordinal indexing the allowability hierarchy.

For a local lower-part model m which carries out the same canonical recursions, write

$$\vec{D}^m, \quad \vec{S}^m, \quad \mathcal{D}^m, \quad \text{and} \quad \varrho_m$$

for its versions of the four objects just defined. The following coherence statement is the form needed in the later model-code localization arguments.

Lemma 4.23. Let $\mathcal{J}_\theta^{M_n}$ be an ω -sound initial segment containing

$$\vec{D}^n, \quad \vec{S}^n, \quad \mathcal{D}^n, \quad \varrho_n,$$

and let

$$X \prec \mathcal{J}_\theta^{M_n}$$

be countable and contain these parameters. Put

$$\lambda = X \cap \omega_1$$

and let $\pi : \bar{X} \rightarrow X$ be the inverse of the transitive collapse. Assume that the hull is chosen as in Theorem 4.17. Then $\bar{X} \trianglelefteq M_n$, $\omega_1^{\bar{X}} = \lambda$, and the collapse of the four global parameters is exactly the corresponding system computed in $\bar{X}$. In particular,

$$\vec{D}^{\bar{X}} = \vec{D}^n \upharpoonright \lambda, \quad \vec{S}^{\bar{X}} = \vec{S}^n \upharpoonright \lambda,$$

$$\mathcal{D}^{\bar{X}} = \langle d_\xi^n \mid \xi < \lambda \rangle,$$

and, for every $s \in (<^\lambda 2)^{\bar{X}}$,

$$\varrho_{\bar{X}}(s) = \varrho_n(s)$$

whenever the objects are identified through the collapse.

Proof. By Theorem 4.17, the transitive collapse $\bar{X}$ is an initial segment of M_n . The definitions of the diamond sequence, the simultaneous Suslin-tree sequence, the almost disjoint family, and the area map are all first-order definitions from the canonical construction order. Elementarity therefore sends each global object to the object obtained by carrying out the same definition in the collapse. Since $\text{crit}(\pi) = \lambda$, the collapse fixes every ordinal below λ and every subset of such an ordinal which belongs to $\bar{X}$. The displayed identities follow. $\square$

Corollary 4.24. *Let $N = M_n[G]$ be a small M_n -generic extension. From the unique set $M_n|_{\omega_1}$ satisfying Θ_n , the model N uniformly recovers*

$$\bar{D}^n, \quad \vec{S}^n, \quad \mathcal{D}^n, \quad \text{and} \quad \varrho_n.$$

Moreover, whenever a countable elementary hull is taken in the later localization argument, its transitive collapse computes the corresponding local versions of all four objects, as described in Lemma 4.23.

Proof. The recovery of $M_n|_{\omega_1}$ is Lemma 4.14. The uniform definitions of the four parameters were established above. The final assertion is exactly the condensation-coherence lemma. $\square$

4.4 The preparatory branch extension W_n

We now pass to the ground model over which the coding iteration will be performed. Throughout this subsection, ω_1 denotes $\omega_1^{M_n}$. Let

$$\vec{S}^n = \langle S_\xi^n \mid \xi < \omega_1 \rangle$$

be the canonical independent sequence of Suslin trees from Lemma 4.20. All products in this subsection are computed in M_n .

Definition 4.25. *For $A \subseteq \omega_1$ with $A \in M_n$, put*

$$\text{Br}_A^n := \prod_{\xi \in A}^{\text{fs}} S_\xi^n.$$

We write

$$\text{Br}_n := \text{Br}_{\omega_1}^n = \prod_{\xi < \omega_1}^{\text{fs}} S_\xi^n.$$

Let $H \subseteq \text{Br}_n$ be M_n -generic. For $\xi < \omega_1$, define

$$b_\xi := \bigcup \{p(\xi) : p \in H \text{ and } \xi \in \text{dom}(p)\}.$$

For $A \subseteq \omega_1$ in M_n , let

$$H_A := H \cap \text{Br}_A^n.$$

Finally, put

$$W_n := M_n[H].$$

Thus b_ξ is the branch added through S_ξ^n by the ξ -th coordinate. As before, to *destroy* a tree S_ξ^n means to add a cofinal generic branch through it.

Lemma 4.26. *Let $A, B \in M_n$ satisfy*

$$A \subseteq B \subseteq \omega_1.$$

Then the following hold in M_n .

1. Br_A^n is c.c.c.
2. Br_A^n is a complete subforcing of Br_B^n , and

$$\text{Br}_B^n \cong \text{Br}_A^n \times \text{Br}_{B \setminus A}^n.$$

3. If $\xi \notin A$, then

$$\text{Br}_A^n \Vdash "S_\xi^n \text{ is a Suslin tree}."$$

Proof. The sequence $\vec{S}^n$ is independent in M_n , so every nonempty finite product of its members is a Suslin tree. Apply Lemma 2.2 inside M_n . The second clause is the coordinatewise factorization of the finite-support product. For the last clause, note that

$$\text{Br}_A^n \times S_\xi^n \cong \text{Br}_{A \cup \{\xi\}}^n,$$

which is c.c.c. by the first clause. The product characterization of preservation of a Suslin tree therefore gives the conclusion. $\square$

Theorem 4.27. *The model $W_n = M_n[H]$ has the following properties.*

1. *The forcing Br_n is c.c.c. and has size $\aleph_1$ in M_n . Consequently, W_n preserves every cardinal and cofinality of M_n , and*

$$W_n \models \text{CH}.$$

2. *Every S_ξ^n is destroyed in W_n , as witnessed by the branch b_ξ .*
3. *If $A \subseteq \omega_1$ belongs to M_n , then H_A is Br_A^n -generic over M_n , and, for every $\xi < \omega_1$,*

$$M_n[H_A] \models "S_\xi^n \text{ has a cofinal branch}" \iff \xi \in A.$$

4. *Every real in W_n belongs to $M_n[H_A]$ for some countable set $A \subseteq \omega_1$ with $A \in M_n$.*
5. *More generally, every member of $H(\omega_1)^{W_n}$ belongs to $M_n[H_A]$ for some countable set $A \subseteq \omega_1$ with $A \in M_n$.*

Proof. This is the argument of Theorem 2.7, carried out inside M_n , with $\vec{S}$, Br , and L replaced by $\vec{S}^n$, Br_n , and M_n , respectively. $\square$

The last two clauses are the branch-coordinate localization needed by the coding argument. We record a form in which a prescribed countable set of coordinates may simultaneously be retained.

Corollary 4.28. *Let $a \in H(\omega_1)^{W_n}$, and let $C \in [\omega_1]^\omega \cap M_n$. Then there is $A \in [\omega_1]^\omega \cap M_n$ such that*

$$C \subseteq A, \quad a \in M_n[H_A],$$

and

$$M_n[H_A] \models "S_\xi^n \text{ is a Suslin tree}" \quad (\xi \in \omega_1 \setminus A).$$

In particular, no cofinal branch through any S_ξ^n with $\xi \notin A$ belongs to $M_n[H_A]$.

Proof. By Theorem 4.27(5), choose a countable branch support $A_0 \in M_n$ for a , and put

$$A = A_0 \cup C.$$

Then A is countable and belongs to M_n . Since $\text{Br}_{A_0}^n$ is a complete subforcing of Br_A^n , we have $a \in M_n[H_A]$. The assertion about the unused trees follows from Lemma 4.26(3). $\square$

Corollary 4.29. *Let $A \subseteq \omega_1$ belong to M_n . Then $M_n[H_A]$, and in particular W_n , is a small M_n -generic extension in the sense of Definition 4.5. Consequently, $M_n[H_A]$ uniformly recovers the true objects*

$$M_n|_{\omega_1}, \quad \vec{D}^n, \quad \vec{S}^n, \quad \mathcal{D}^n, \quad \varrho_n.$$

The recovered objects are independent of A and agree with the objects fixed in M_n .

Proof. By Lemma 4.26, the forcing Br_A^n is c.c.c., cardinal preserving, and has size at most $\aleph_1$ in M_n . It is therefore a small forcing in the sense of Definition 4.5. Apply Lemma 4.14 and Corollary 4.24. $\square$

From now on, W_n is the ground model of the coding and uniformization iterations. The full branch family

$$\langle b_\xi \mid \xi < \omega_1 \rangle$$

is available in W_n , while Theorem 4.27 and Corollary 4.28 allow any countable collection of parameters to be localized to an inner branch extension which omits all but countably many of these branches. This tension between global availability and local omission is the mechanism used below to rule out unintended codes.

4.5 The adapted M_n -coding mechanism

We now adapt the coding forcing from Section 2 to the branch model W_n . The canonical objects used in the definition are

$$M_n|_{\omega_1}, \quad \vec{S}^n, \quad \mathcal{D}^n, \quad \varrho_n,$$

and, by Corollary 4.29, they are computed uniformly and correctly in every branch subextension which occurs in the localization arguments. Throughout this subsection, ω_1 denotes the common first uncountable cardinal of M_n and W_n .

The reservoir of coding areas

Definition 4.30. *Put*

$$\mathbb{C}_{M_n}(\omega_1) := ({}^{<\omega_1}2)^{M_n},$$

ordered by end-extension. Thus a condition is a countable binary sequence belonging to M_n . We use a tagged copy $\mathbb{C}_{M_n}(\omega_1)_\eta$ for each $\eta < \omega_1$ and put

$$\mathbb{C}_n^{\text{area}} := \prod_{\eta < \omega_1}^{\text{cs}} \mathbb{C}_{M_n}(\omega_1)_\eta.$$

For $E \subseteq \omega_1$, let

$$(\mathbb{C}_n^{\text{area}})_E := \prod_{\eta \in E}^{\text{cs}} \mathbb{C}_{M_n}(\omega_1)_\eta.$$

The canonical name for the generic subset of ω_1 at coordinate η is denoted by $\dot{g}_\eta$.

The product presentation is retained because its coordinates supply the coding areas used by the later finite-support iteration. As a forcing, the reservoir is equivalent to a single copy of the M_n -computed ω_1 -Cohen forcing.

Lemma 4.31. *The following hold.*

1. *In M_n , the forcing $\mathbb{C}_n^{\text{area}}$ is σ -closed and has size $\aleph_1$.*
2. *There is an M_n -definable isomorphism*

$$\mathbb{C}_n^{\text{area}} \cong \mathbb{C}_{M_n}(\omega_1).$$

The same holds for every product on a set of $\aleph_1$ many coordinates and after every permutation of the coordinates which belongs to M_n .

3. *In W_n the forcing $\mathbb{C}_n^{\text{area}}$ is ω -distributive. Consequently it adds no reals over W_n , preserves ω_1 , and preserves CH.*

4. If $A, E \subseteq \omega_1$ belong to M_n , then the forcing $(\mathbb{C}_n^{\text{area}})_E$ may be commuted in front of Br_A^n . In the resulting reservoir extension, every tree S_ξ^n which was Suslin before the branch forcing remains Suslin.

Proof. This is the reservoir argument from Subsection 2.4, carried out with M_n in place of L . $\square$

Definition 4.32. Let $g \subseteq \omega_1$ be $\mathbb{C}_{M_n}(\omega_1)$ -generic over a model containing the canonical area map ϱ_n . The coding area determined by g is

$$h(g) := \{\varrho_n(g \restriction \xi) : \xi < \omega_1\}.$$

If $K \subseteq \mathbb{C}_n^{\text{area}}$ is generic, we write

$$g_\eta := \dot{g}_\eta^K, \quad h_\eta := h(g_\eta)$$

for the generic and the coding area at coordinate η .

Every proper initial segment of g belongs to M_n . Moreover, since the forcing is σ -closed in M_n , for every $\alpha < \omega_1$ the set $h(g) \cap \alpha$ belongs to M_n . Thus a coding area is a new unbounded subset of ω_1 all of whose bounded initial segments are already available in the canonical lower part.

Lemma 4.33. Let g and g' be distinct M_n -Cohen subsets of ω_1 . Then

$$h(g) \cap h(g')$$

is bounded in ω_1 . In particular, the family

$$\langle h_\eta \mid \eta < \omega_1 \rangle$$

provided by a reservoir generic is pairwise almost disjoint modulo bounded sets.

Proof. This is the same separation argument as in the L -construction, with the area map ϱ_L replaced by ϱ_n . $\square$

We retain the notation introduced in Section 2. For $u \in 2^\omega$, $\gamma < \omega_1$, and $k < \omega$, put

$$\tau_u(\gamma, k) = \begin{cases} \omega\gamma + 2k + 1, & k \in u, \\ \omega\gamma + 2k, & k \notin u, \end{cases}$$

and, for $h \subseteq \omega_1$, put

$$\text{Sel}(h, u) := \{\tau_u(\gamma, k) : \gamma \in h, k < \omega\}.$$

Thus the k -th bit of u is written at the γ -th block by selecting one of the two trees

$$S_{\omega\gamma+2k}^n, \quad S_{\omega\gamma+2k+1}^n.$$

The following elementary strengthening of Lemma 4.33 is the combinatorial form used in the later no-unwanted-codes argument.

Lemma 4.34. *Let I be countable, let $\langle h_i \mid i \in I \rangle$ be coding areas obtained from distinct reservoir coordinates, let $u_i \in 2^\omega$, and let $B \in [\omega_1]^\omega$. Put*

$$U := B \cup \bigcup_{i \in I} \text{Sel}(h_i, u_i).$$

Then:

1. *If $h \subseteq \omega_1$ has size ω_1 and $h \cap h_i$ is bounded for every $i \in I$, then there are ω_1 many $\gamma \in h$ such that*

$$[\omega\gamma, \omega\gamma + \omega) \cap U = \emptyset.$$

2. *Suppose that $i_0 \in I$, that $h \cap h_{i_0}$ is unbounded, and that $u \neq u_{i_0}$. If $k < \omega$ satisfies $u(k) \neq u_{i_0}(k)$, then there are ω_1 many $\gamma \in h \cap h_{i_0}$ such that*

$$\tau_u(\gamma, k) \notin U.$$

Proof. This is the countable version of the preceding separation argument, exactly as in the L -case. By Lemma 4.33, distinct coding areas have bounded intersection. Since I is countable, one bound controls all pairwise intersections among the areas under consideration. Also exclude the countable set of blocks meeting B .

In case (1), any sufficiently large $\gamma \in h$ outside these exceptional sets has its whole ω -block disjoint from U . In case (2), choose such a sufficiently large $\gamma \in h \cap h_{i_0}$; the inequality $u(k) \neq u_{i_0}(k)$ means that, at the k -th pair of the γ -block, u selects the tree opposite to the one selected by u_{i_0} . Thus $\tau_u(\gamma, k) \notin U$. In both cases there are ω_1 many possible choices of γ . $\square$

Writing a real into the branch sequence

Let $u \in 2^\omega$ belong to a forcing extension of W_n , let g be a fresh M_n -Cohen generic, and put $h = h(g)$. The selected branch sequence is

$$\mathcal{B}_{g,u}^n := \langle b_{\tau_u(\gamma,k)} \mid \gamma \in h, k < \omega \rangle.$$

We regard this as an indexed set of triples

$$(\gamma, k, b_{\tau_u(\gamma,k)}).$$

Consequently the index and the tree through which the displayed branch runs recover the bit $u(k)$. This formulation uses the distinguished selected branch sequence; it does not make any assertion about branches through the opposite member of the pair in the ambient universe.

Put

$$m_\infty^n := M_n \restriction \omega_1.$$

Choose canonically a transitive model

$$\mathcal{N}_{g,u}$$

of size $\aleph_1$ such that:

1. $\mathcal{N}_{g,u} \models \text{ZFC}^- + \text{"}\aleph_1 \text{ exists"}$ and $\omega_1^{\mathcal{N}_{g,u}} = \omega_1$;
2. m_∞^n , h , and $\mathcal{B}_{g,u}^n$ belong to $\mathcal{N}_{g,u}$;
3. using m_∞^n , the model $\mathcal{N}_{g,u}$ computes $\vec{S}^n$, $\mathcal{D}^n$, and ϱ_n correctly;
4. $\mathcal{N}_{g,u}$ verifies that every displayed member of $\mathcal{B}_{g,u}^n$ is a cofinal branch through the corresponding tree and reads the same bit of u at every block whose starting point belongs to h .

For definiteness, take the least model of the form

$$L_\theta[m_\infty^n, h, \mathcal{B}_{g,u}^n]$$

which satisfies the displayed theory and contains the indicated objects. The usual reflection argument gives such a θ of cardinality $\aleph_1$. This choice supplies a canonical constructibility order and therefore a canonical enumeration of $\mathcal{N}_{g,u}$ of length ω_1 .

Let

$$X_{g,u} \subseteq \omega_1$$

be the resulting canonical code for

$$(\mathcal{N}_{g,u}, \in, m_\infty^n, h, \mathcal{B}_{g,u}^n).$$

The code records a well-founded extensional relation on ω_1 together with constants for the three displayed objects. Decoding $X_{g,u}$ and taking the transitive collapse therefore recovers the entire tuple.

Localization of the model code

We now apply to $X_{g,u}$ the localization procedure introduced in the L -case.

For $Y \subseteq \omega_1$, put

$$O(Y) := \{\alpha < \omega_1 : 2\alpha + 1 \in Y\}, \quad E(Y) := \{\alpha < \omega_1 : 2\alpha \in Y\}.$$

For every countable ordinal α , let $\text{wo}_n(\alpha)$ be the $<_{M_n}$ -least real coding a wellorder of ω of order type α .

Choose canonically an ordinal λ of size $\aleph_1$ such that

$$L_\lambda[X_{g,u}] \models \text{ZFC}^- + \text{"}\aleph_1 \text{ exists"},$$

and, inside this model, choose the canonical continuous increasing chain

$$\langle M_\xi \mid \xi < \omega_1 \rangle$$

of countable elementary submodels whose union is $L_\lambda[X_{g,u}]$. We require that all decoding parameters belong to M_0 and that, putting

$$c_\xi := M_\xi \cap \omega_1,$$

one has

$$c_0 > \omega, \quad c_{\xi+1} > c_\xi + \omega.$$

Let

$$C_{g,u} := \{c_\xi : \xi < \omega_1\}.$$

Definition 4.35. *The localization of $X_{g,u}$ is the unique set*

$$Y_{g,u} \subseteq \omega_1$$

with

$$O(Y_{g,u}) = X_{g,u}$$

and whose even part is defined as follows:

1. $E(Y_{g,u}) \cap \omega$ is the fixed real $\text{wo}_n(c_0)$;
2. $E(Y_{g,u}) \cap [\omega, c_0) = \emptyset$;
3. for every $\xi < \omega_1$, $E(Y_{g,u}) \cap [c_\xi, c_\xi + \omega)$ is the translate of $\text{wo}_n(c_{\xi+1})$ to that interval;
4. for every $\xi < \omega_1$,

$$E(Y_{g,u}) \cap [c_\xi + \omega, c_{\xi+1}) = \emptyset.$$

We also write

$$Y_n(u, g) := Y_{g,u}.$$

All choices in the construction of $\mathcal{N}_{g,u}$, $X_{g,u}$, the hull sequence, and $Y_{g,u}$ are made by the canonical orders fixed above. Hence, for forcing names $\dot{u}$ and $\dot{g}$, the notation

$$\dot{Y}_n(\dot{u}, \dot{g})$$

denotes a canonical forcing name.

Lemma 4.36. *Let M be a countable transitive model of $\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$ and put $\alpha = \omega_1^M$. Suppose that*

$$Y_{g,u} \cap \alpha \in M.$$

Then $\alpha \in C_{g,u}$. Moreover, inside its version of $L[Y_{g,u} \cap \alpha]$, the model M recovers

$$X_{g,u} \cap \alpha$$

and the well-founded extensional structure coded by this initial segment. Its transitive collapse is the local structure obtained by collapsing the corresponding elementary hull M_ξ with $c_\xi = \alpha$. In particular, the local collapse contains the appropriate initial segment of m_∞^n , the local coding area, and the local selected branch sequence.

Proof. This is the same localization argument as in the L -case. The odd coordinates of $Y_{g,u} \cap \alpha$ give $X_{g,u} \cap \alpha$. The even coordinates force $\alpha \in C_{g,u}$: if $\alpha < c_0$, or if $c_\xi < \alpha < c_{\xi+1}$, then, in the latter case using $c_\xi + \omega < \alpha$, the model M contains the real $\text{wo}_n(c_0)$, respectively $\text{wo}_n(c_{\xi+1})$, and hence a well-order of ω of type at least $\alpha = \omega_1^M$, contradiction. Thus $\alpha = c_\xi$ for some ξ . By the choice of the elementary hull chain, $X_{g,u} \cap c_\xi$ is exactly the code of the collapse of M_ξ . The distinguished constants collapse to the corresponding local initial segment of m_∞^n , the local coding area, and the local selected branch sequence. The remaining assertions follow from uniqueness of the Mostowski collapse and from Lemma 4.23. $\square$

Almost disjoint coding and the forcing Code_n

For $Y \subseteq \omega_1$, let

$$\mathbb{A}^n(Y) := \mathbb{A}_{\mathcal{D}^n}(Y)$$

be the Jensen–Solovay almost disjoint coding forcing relative to the canonical family

$$\mathcal{D}^n = \langle d_\xi^n \mid \xi < \omega_1 \rangle.$$

Thus, if $G \subseteq \mathbb{A}^n(Y)$ is generic and

$$r_G := \bigcup \{s : \exists R ((s, R) \in G)\},$$

then

$$\xi \in Y \iff r_G \cap d_\xi^n \text{ is finite}$$

for every $\xi < \omega_1$.

Definition 4.37. Let $u \in 2^\omega$ and let $\eta < \omega_1$. The tagged two-step coding forcing is

$$\text{Code}_n(u, \eta) := \mathbb{C}_{M_n}(\omega_1)_\eta * \dot{\mathbb{A}}^n(\dot{Y}_n(\check{u}, \dot{g}_\eta)).$$

If u is given by a name $\dot{u}$, we use the corresponding name

$$\text{Code}_n(\dot{u}, \eta) := \mathbb{C}_{M_n}(\omega_1)_\eta * \dot{\mathbb{A}}^n(\dot{Y}_n(\dot{u}, \dot{g}_\eta)).$$

The ordinal η merely tags the copy of the Cohen forcing in the standalone definition. Its real role appears in an iteration over the global reservoir. Once $K \subseteq \mathbb{C}_n^{\text{area}}$ has been added, the coordinate g_η is already present, and a coding stage using the fresh area η consists only of the c.c.c. forcing

$$\mathbb{A}^n(Y_n(u, g_\eta)).$$

Thus, exactly as in the L -construction, the Cohen reservoir is added once, while only the almost disjoint coding forcings are iterated with finite support.

Lemma 4.38. *Let $u \in 2^\omega$ belong to the model over which the coding is performed. Then:*

1. *the reservoir step adds no real over W_n ;*
2. *in the reservoir extension,*

$$\mathbb{A}^n(Y_n(u, g_\eta))$$

is Knaster and has size $\aleph_1$;

3. *$\text{Code}_n(u, \eta)$ preserves ω_1 and CH;*
4. *the generic real r added by the second step canonically decodes $Y_n(u, g_\eta)$ and hence the localized code of*

$$(\mathcal{N}_{g_\eta, u}, \in, m_\infty^n, h_\eta, \mathcal{B}_{g_\eta, u}^n);$$

5. *the branch coordinates occurring in the construction of the localized code are exactly*

$$\text{Sel}(h_\eta, u).$$

Proof. Clause (1) is Lemma 4.31(3). Clause (2) follows from Lemma 2.4, applied to the canonical almost disjoint family $\mathcal{D}^n$. The forcing has size $\aleph_1$ because $\mathcal{D}^n$ has length ω_1 .

The first step preserves ω_1 and CH, and the second step is c.c.c. of size $\aleph_1$; hence (3) follows. Clause (4) is the defining property of Jensen–Solovay almost disjoint coding together with the fixed localization decoding procedure. Finally, $Y_n(u, g_\eta)$ depends on the branch family only through $\mathcal{B}_{g_\eta, u}^n$, whose members are precisely the branches indexed by $\text{Sel}(h_\eta, u)$. This proves (5). $\square$

4.6 The projective coding predicates and exactness

We now isolate the projective content of the forcing Code_n .

Lemma 4.39. *Let N be a small M_n -generic extension, let $m \in \mathcal{I}_n^N$, and put*

$$\alpha := \omega_1^m.$$

Then the canonical objects computed in m agree with the corresponding initial parts of the global objects. In particular,

$$\vec{S}^m = \vec{S}^n \upharpoonright \alpha, \quad \mathcal{D}^m = \langle d_\xi^m \mid \xi < \alpha \rangle,$$

and, under the canonical identification of the domains,

$$\varrho_m(s) = \varrho_n(s)$$

for every $s \in ({}^{<\alpha}2)^m$.

Proof. By Lemma 4.12, the premouse m is a proper initial segment of M_n . The canonical parameters are obtained by first-order recursions from the construction order of M_n , and the restriction of that order to m is the construction order computed by m . The uniqueness of the recursions therefore gives the displayed identities. Equivalently, one may take a countable hull containing the relevant global parameters and apply Lemma 4.23. $\square$

The local decoding predicate

Let M be a transitive model of $\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$, let $m \in M$ be a premouse, and suppose that

$$\omega_1^m = \omega_1^M =: \alpha.$$

Inside M , the notation

$$\text{LocalCode}_n(Y, u, m)$$

denotes the following first-order assertion about $Y \subseteq \alpha$ and $u \in 2^\omega$.

First, Y is a correctly localized code in the sense of Definition 4.35. Its odd part decodes a set $X \subseteq \alpha$, its even part supplies the corresponding localization data, and X codes a well-founded extensional structure with distinguished constants. The transitive collapse of that structure is a model $\bar{\mathcal{N}}$ of cardinality $\aleph_1^M$ such that

$$\bar{\mathcal{N}} \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}, \quad \omega_1^{\bar{\mathcal{N}}} = \alpha.$$

The distinguished constants decode:

1. a lower-part premouse m^* with $\omega_1^{m^*} = \alpha$;
2. an unbounded coding area $h \subseteq \alpha$;
3. an indexed branch sequence

$$\mathcal{B} = \langle c_{\gamma, k} \mid \gamma \in h, k < \omega \rangle.$$

The inverse images of the members of h under ϱ_{m^*} form an end-extension chain. Its union is a sequence $g \in {}^\alpha 2$ such that

$$\forall \xi < \alpha (g \restriction \xi \in m^*), \quad h = \{\varrho_{m^*}(g \restriction \xi) : \xi < \alpha\}.$$

The premice m and m^* compute the same local canonical parameters

$$\vec{S}^m, \quad \mathcal{D}^m, \quad \varrho_m.$$

Finally, for every $\gamma \in h$ and $k < \omega$, the object $c_{\gamma,k}$ is a cofinal branch through

$$S_{\tau_u(\gamma,k)}^m.$$

Thus the real obtained from the parity of the tree indices in the distinguished branch sequence is exactly u . We make no assertion about branches through the opposite members of the pairs in the ambient universe; in W_n every tree in $\vec{S}^n$ already has a branch.

All clauses in this description are first-order over M . In particular, LocalCode_n does not quantify over an iteration strategy. The projective complexity enters only through the external requirement that m belong to $\mathcal{I}_n$.

Let

$$\mathcal{D}^m = \langle d_\xi^m \mid \xi < \omega_1^m \rangle$$

be the almost disjoint family computed in m .

Definition 4.40. For reals r, u , let $\Psi_n(r, u)$ be the assertion

$$\begin{aligned} & \forall M \forall m \left(M \text{ is a countable transitive model of } \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”} \right. \\ & \quad \wedge \ r, u, m \in M \\ & \quad \wedge \ m \in \mathcal{I}_n \\ & \quad \wedge \ \omega_1^m = \omega_1^M \\ & \quad \implies \exists Y \in M \left[Y \subseteq \omega_1^M \right. \\ & \quad \quad \wedge \ \forall \xi < \omega_1^M \left(\xi \in Y \iff r \cap d_\xi^m \text{ is finite} \right) \\ & \quad \quad \wedge \ M \models \text{LocalCode}_n(Y, u, m) \left. \right] \Big). \end{aligned}$$

We then put

$$\Phi_n(u) \iff \exists r \Psi_n(r, u).$$

When $\Phi_n(u)$ holds, we say that u is coded into $\vec{S}^n$, or that u is written into $\vec{S}^n$.

Lemma 4.41. The predicate $\Psi_n(r, u)$ is Π_{n+2}^1 , uniformly in r, u , and $\Phi_n(u)$ is Σ_{n+3}^1 .

Proof. We use reals to code the countable structures M and m . The assertions that a real codes a well-founded countable transitive model of the displayed fragment of set theory, that the indicated objects belong to it, and that it satisfies the fixed first-order formula LocalCode_n have complexity at most Π_1^1 , or are arithmetic once well-foundedness has been assumed. By Lemma 4.12, the assertion $m \in \mathcal{I}_n$ is Π_{n+1}^1 . The existential quantifier over $Y \in M$ is a quantifier over an element of the coded countable structure and hence is arithmetic in its code.

After rewriting the implication as a disjunction, the matrix following the two universal real quantifiers is Σ_{n+1}^1 . Therefore Ψ_n is Π_{n+2}^1 . Adding the existential real quantifier for r gives $\Phi_n \in \Sigma_{n+3}^1$. $\square$

The global objects decoded by a witness

Lemma 4.42. *Work in a small M_n -generic extension and suppose that $\Psi_n(r, u)$ holds. Put*

$$m_\infty^n = M_n|_{\omega_1}.$$

Then the inner model $L[r, m_\infty^n]$ contains u and canonically decodes objects

$$Y_r, \quad X_r, \quad \mathcal{N}_r, \quad g_r, \quad h_r, \quad \mathcal{B}_r$$

with the following properties:

1.

$$Y_r = \{\xi < \omega_1 : r \cap d_\xi^n \text{ is finite}\};$$

the odd part of Y_r decodes X_r , and $\mathcal{N}_r$ is the transitive collapse of the structure coded by X_r ;

2. $g_r \in {}^{\omega_1}2$, every proper initial segment of g_r belongs to m_∞^n , and

$$h_r = \{\varrho_n(g_r \upharpoonright \xi) : \xi < \omega_1\};$$

in particular, $|h_r| = \omega_1$;

3.

$$\mathcal{B}_r = \langle c_{\gamma, k}^r \mid \gamma \in h_r, k < \omega \rangle$$

is an indexed family of actual cofinal branches and

$$c_{\gamma, k}^r \text{ is a branch through } S_{\tau_u(\gamma, k)}^n$$

for every $\gamma \in h_r$ and $k < \omega$.

Consequently,

$$\text{Sel}(h_r, u) \subseteq \{\xi < \omega_1 : L[r, m_\infty^n] \text{ contains a cofinal branch through } S_\xi^n\}.$$

Proof. The almost disjoint decoding of r relative to the canonical family $\mathcal{D}^n$ is unique, so $L[r, m_\infty^n]$ can form the displayed set Y_r . We claim that $L[r, m_\infty^n]$ can canonically decode from Y_r the objects

$$X_r, \quad \mathcal{N}_r, \quad g_r, \quad h_r, \quad \mathcal{B}_r$$

appearing in the statement of the lemma.

Suppose otherwise. Then a part of the first-order decoding fails: the localization test fails, the odd part fails to code the required well-founded extensional structure, one of the distinguished constants is missing, or one of the branch clauses fails. Choose a sufficiently large regular Θ and a countable elementary submodel

$$Z \prec H_\Theta$$

which contains r, u, m_∞^n and a witness to the failure. We choose Z from the standard club of lower-part hulls for which the collapse of the lower-part parameter belongs to $\mathcal{I}_n$. Let

$$\pi : Z \longrightarrow \bar{M}$$

be the transitive collapse and put

$$m := \pi(m_\infty^n).$$

By Lemmas 4.12 and 4.23,

$$m \in \mathcal{I}_n, \quad \omega_1^m = \omega_1^{\bar{M}},$$

and m computes the correct initial parts of $\vec{S}^n$, $\mathcal{D}^n$, and ϱ_n .

Inside $\bar{M}$, the almost disjoint decoding of r relative to $\mathcal{D}^m$ is exactly

$$Y_r \cap \omega_1^{\bar{M}}.$$

Indeed, for every $\xi < \omega_1^{\bar{M}}$, Lemma 4.39 gives $d_\xi^m = d_\xi^n$. The failure of the global decoding reflects to $\bar{M}$, and hence

$$\bar{M} \not\models \text{LocalCode}_n(Y_r \cap \omega_1^{\bar{M}}, u, m).$$

On the other hand, $\Psi_n(r, u)$ applies to $(\bar{M}, m)$ and produces a set $Y \in \bar{M}$ which is almost-disjointly decoded from r and satisfies

$$\bar{M} \models \text{LocalCode}_n(Y, u, m).$$

The almost disjoint decoding is unique, so

$$Y = Y_r \cap \omega_1^{\bar{M}},$$

a contradiction. Thus the global decoding succeeds.

Let u^* be the real read in $L[r, m_\infty^n]$ from the parity of the tree indices in the decoded branch sequence. If $u^* \neq u$, choose $k < \omega$ on which they differ and repeat the preceding countable-hull argument, now including enough of the decoded data to determine $u^*(k)$. The unique local decoding would read the same indexed branch family as coding both $u^*(k)$ and $u(k)$, which is impossible. Hence $u^* = u$, so $u \in L[r, m_\infty^n]$.

The inverse images under ϱ_n of the members of the decoded area form an end-extension chain of proper initial segments. Their union is g_r , and injectivity of ϱ_n gives $|h_r| = \omega_1$. The remaining conclusions follow directly from the successful global decoding. $\square$

Theorem 4.43. *Let N be a small M_n -generic extension containing a real u , and let r be the real added by $\text{Code}_n(u, \eta)$ over N . Let g_η denote the Cohen generic added by the first step. Then*

$$N[g_\eta][r] \models \Psi_n(r, u),$$

and hence

$$N[g_\eta][r] \models \Phi_n(u).$$

Moreover, the same real r witnesses $\Psi_n(r, u)$ in every later small M_n -generic extension of $N[g_\eta][r]$ occurring in the construction.

Proof. Let g_η be the Cohen generic at the first step and put

$$Y = Y_n(u, g_\eta).$$

The real r added at the second step almost-disjointly codes Y relative to $\mathcal{D}^n$.

Work in $N[g_\eta][r]$ and let M, m satisfy the antecedent in the definition of $\Psi_n(r, u)$. Put $\alpha = \omega_1^M$. By Lemma 4.39, m computes the initial almost disjoint family and the remaining canonical parameters correctly. Therefore the set decoded from r inside M is

$$Y \cap \alpha.$$

Lemma 4.36, together with condensation coherence, shows that M decodes from this set the corresponding local collapse of the model coded by Y . Its distinguished area and branch sequence satisfy the u -pattern. Hence

$$M \models \text{LocalCode}_n(Y \cap \alpha, u, m).$$

This proves $\Psi_n(r, u)$.

The formula $\Psi_n(r, u)$ is Π_{n+2}^1 by Lemma 4.41. The persistence assertion therefore follows from Theorem 4.9, applied to the relevant intermediate and later small forcing extensions. The witness r is unchanged. $\square$

Coding iterations and exactness

The exactness argument is most conveniently proved already for the basic coding iterations which will form level zero of the allowability hierarchy.

Definition 4.44. *Let $\delta \leq \omega_1$. An M_n -coding iteration over the reservoir of length δ consists of an injective sequence*

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle \in M_n$$

and a sequence of names

$$\langle \dot{u}_\beta \mid \beta < \delta \rangle$$

for reals. Put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

Recursively define

$$\mathbb{P}_\beta = \mathbb{C}_n^{\text{area}} * \dot{\mathbb{R}}_\beta \quad (\beta \leq \delta)$$

as follows. The forcing $\dot{\mathbb{R}}_0$ is trivial. At stage β , the name $\dot{u}_\beta$ is required to be the canonical lift of a name over

$$(\mathbb{C}_n^{\text{area}})_{U_\beta} * \dot{\mathbb{R}}_\beta,$$

and

$$\dot{\mathbb{R}}_{\beta+1} = \dot{\mathbb{R}}_\beta * \dot{\mathbb{A}}^n(\dot{Y}_n(\dot{u}_\beta, \dot{g}_{\eta_\beta})).$$

*At limit stages, $\dot{\mathbb{R}}_\lambda$ is the finite-support limit. If $K * G$ is generic for $\mathbb{P}_\delta$, put*

$$u_\beta := \dot{u}_\beta^{K*G}, \quad h_\beta := h(g_{\eta_\beta}).$$

The reals u_β , $\beta < \delta$, are called the reals intentionally coded by the iteration.

Lemma 4.45. *Every M_n -coding iteration of length at most ω_1 has size $\aleph_1$, preserves cardinals, and preserves CH. Over the reservoir extension, its almost disjoint tail is c.c.c.*

Proof. The reservoir has size $\aleph_1$, is ω -distributive over W_n , and preserves CH by Lemma 4.31. Every tail factor is Knaster and has size $\aleph_1$ by Lemma 4.38. A finite-support iteration of at most ω_1 many c.c.c. forcings is c.c.c.; under CH the iteration has size $\aleph_1$. The asserted preservation follows. $\square$

For $\xi < \omega_1$, put

$$W_{n,\xi} := M_n[H_{\omega_1 \setminus \{\xi\}}].$$

This is a genuine branch subextension defined by a ground-model product. It contains every branch b_ζ with $\zeta \neq \xi$, while S_ξ^n remains Suslin in $W_{n,\xi}$.

Lemma 4.46. *Let $\mathbb{P}_\delta$ be an M_n -coding iteration over the reservoir, let $K * G$ be generic over W_n , and let $\vec{x}$ be a finite tuple of reals in $W_n[K][G]$. There are countable sets*

$$E \subseteq \omega_1, \quad I \subseteq \delta, \quad B \subseteq \omega_1,$$

with $B \in M_n$, such that:

1. I is recursively closed in the following sense: whenever $\beta \in I$, the name $\dot{u}_\beta$ defining the β -th iterand is a name over the induced closed subiteration on $I \cap \beta$ together with the reservoir coordinates in E , and $\eta_\beta \in E$.

2. the tuple $\vec{x}$ is read by the induced closed subiteration on I together with the reservoir coordinates in E ;

3. for every

$$\xi \notin B \cup \bigcup_{\beta \in I} \text{Sel}(h_\beta, u_\beta),$$

the induced forcing on (E, I) is computed in the same way over $W_{n,\xi}$ as over W_n , and

$$\vec{x} \in W_{n,\xi}[K \restriction E][G \restriction I];$$

4. for every such ξ ,

$$W_{n,\xi}[K \restriction E][G \restriction I] \models "S_\xi^n \text{ is a Suslin tree}."$$

Here $G \restriction I$ denotes the generic sequence for the induced closed subiteration.

Proof. This is the proof of Lemma 2.11, with L , $\vec{S}$, D , and ϱ_L replaced by M_n , $\vec{S}^n$, $\mathcal{D}^n$, and ϱ_n .

We recall the points where the M_n -setup enters. Nice names have countable reservoir and iteration support, since the reservoir adds no reals and the almost disjoint tail is c.c.c. Closing these supports recursively under the names defining the retained iterands gives countable sets E and I with the displayed closure property, and Theorem 4.27 gives a countable branch support $B \in M_n$ for the ground-model parameters.

If

$$\xi \notin B \cup \bigcup_{\beta \in I} \text{Sel}(h_\beta, u_\beta),$$

then the same induced subiteration on (E, I) is computed over $W_{n,\xi}$: no retained iterand uses the branch b_ξ . The tree S_ξ^n is Suslin in $W_{n,\xi}$ by Lemma 4.26, and each retained almost disjoint factor is Knaster, hence preserves it. Thus the restricted generic interprets the chosen names for $\vec{x}$ in $W_{n,\xi}[K \restriction E][G \restriction I]$, and S_ξ^n remains Suslin there. $\square$

Theorem 4.47. *Let $\mathbb{P}_\delta$ be an M_n -coding iteration over the reservoir, where $\delta \leq \omega_1$, and let $K * G$ be generic over W_n . Then*

$$W_n[K][G] \models \Phi_n(u) \iff u = u_\beta \text{ for some } \beta < \delta.$$

Thus Φ_n recognizes exactly the reals intentionally coded by the iteration.

Proof. This is the no-unwanted-codes argument from the L -case, with Lemma 2.11 replaced by Lemma 4.46 and with the canonical objects $L, \tilde{S}, D, \varrho_L$ replaced by $M_n, \tilde{S}^n, \mathcal{D}^n, \varrho_n$.

If $u = u_\beta$ is intentionally coded at stage β , then the real added by the corresponding almost disjoint coding factor witnesses $\Phi_n(u)$. The persistence of this witness under the remaining forcing follows from the small-forcing projective absoluteness established above.

Conversely, suppose that r witnesses $\Phi_n(u)$ in the final extension and that u is not intentionally coded. Apply Lemma 4.46 to r and u , and take the union of the resulting supports. Thus there are countable sets E, I, B such that r, u belong to the localized extension determined by these supports. As in the L -case, the coding area decoded from r is either almost disjoint from all retained areas h_β with $\beta \in I$, or it is one of them. In the first case choose a fresh block outside all retained areas and outside the countable branch support B ; in the second case use a coordinate on which u differs from the corresponding intentionally coded real. In both cases we obtain

$$\xi = \tau_u(\gamma, k) \notin B \cup \bigcup_{\beta \in I} \text{Sel}(h_\beta, u_\beta).$$

By Lemma 4.46, the tree S_ξ^n remains Suslin in the localized inner model. On the other hand, the witness r decodes a code asserting that the selected branch through S_ξ^n exists. Since r belongs to the localized model, this gives a cofinal branch through S_ξ^n there, a contradiction. Hence u was intentionally coded.

The length- ω_1 case follows, as in the L -case, by observing that nice names for the relevant reals have countable support and therefore appear in a countable initial segment. $\square$

Theorem 4.47 is the exactness statement needed in the uniformization argument. It completes the M_n -specific coding input; the remaining fixed-point construction is transferred from the L -case in the next subsection.

4.7 Transfer of the fixed-point construction

The preceding subsections isolate all points at which the proof over L uses features of the constructible universe. We now transfer the fixed-point construction from Subsection 2.6 and the final iteration from Section 3. We

spell out the translation and the hypotheses used by the transfer, but do not repeat the forcing-theoretic inductions.

Fix a recursive universal enumeration

$$\langle \varphi_m^n(v_0, v_1) \mid m < \omega \rangle$$

of the Π_{n+3}^1 formulas with two free real variables, and put

$$A_m^n := \{(x, y) \in (2^\omega)^2 : \varphi_m^n(x, y)\}.$$

As before, triples (x, y, m) are identified with their fixed recursive real codes.

The translation. Let $\mathbf{t}_n$ denote the following simultaneous translation of the objects used in the L -construction:

$$\begin{array}{ll} L & \mapsto M_n, \\ \text{Br} & \mapsto \text{Br}_n, \\ \mathbb{C}_L(\omega_1) & \mapsto \mathbb{C}_{M_n}(\omega_1), \\ D & \mapsto \mathcal{D}^n, \\ \text{the area map } \varrho_L & \mapsto \varrho_n, \\ \Psi & \mapsto \Psi_n, \\ A_m & \mapsto A_m^n, \end{array} \quad \begin{array}{ll} W & \mapsto W_n, \\ \vec{S} & \mapsto \vec{S}^n, \\ \mathbb{C}^{\text{area}} & \mapsto \mathbb{C}_n^{\text{area}}, \\ \mathbb{A}_D & \mapsto \mathbb{A}^n, \\ \dot{Y}(\dot{x}, \dot{z}, \dot{m}, \dot{g}_\eta) & \mapsto \dot{Y}_n(\langle \dot{x}, \dot{z}, \dot{m} \rangle, \dot{g}_\eta), \\ \Phi & \mapsto \Phi_n, \\ <_L & \mapsto <_n. \end{array}$$

Here $<_n$ is the canonical order on M_n -codes. Every object in $H(\omega_2)^{W_n}$ is represented by its $<_{M_n}$ -least Br_n -name in M_n , and codes are compared in the canonical construction order of M_n . The same convention is used for forcings, complete embeddings, conditions, and names which occur in localized support pairs.

The semantic least-key selection is translated at the same time. If $\mathfrak{a} = (E, A)$ is an admissible support pair in an M_n -coding iteration, write

$$M_{\mathfrak{a}}^n := W_n[K_E][G_A].$$

For $z \in M_{\mathfrak{a}}^n$, where in the application $z = \langle x, y, m \rangle$, a presentation of z below $\mathfrak{a}$ is defined as in the L -case, replacing allowable forcing by M_n -allowable forcing and requiring the unused coding areas to be disjoint from those used by $\mathfrak{a}$. Thus a presentation determines a pointed M_n -allowable presentation

$$d = (\mathbb{P}_{\mathfrak{b}} * \dot{\mathbb{R}}, (1_{\mathbb{P}_{\mathfrak{b}}}, \dot{p}), \dot{\sigma}).$$

Two such pointed presentations are identified if they have a common M_n -allowable refinement below which the pushed-forward names are forced to be equal. Let

$$\text{pcode}_n(d) := \min_{<_n} \{\text{code}(e) : e \equiv_{\text{pres}}^n d\}.$$

Then

$$\mathcal{O}_{\mathfrak{a}}^n(z) := \{\text{pcode}_n(d_c) : c \text{ is a presentation of } z \text{ below } \mathfrak{a}\}$$

and

$$\text{key}_{\mathfrak{a}}^n(z) := \min_{\prec_n} \mathcal{O}_{\mathfrak{a}}^n(z).$$

The proof of Lemma 2.13 translates verbatim: if $\mathfrak{b} \preccurlyeq \mathfrak{a}$ and $z \in M_{\mathfrak{b}}^n$, then

$$\mathcal{O}_{\mathfrak{b}}^n(z) = \mathcal{O}_{\mathfrak{a}}^n(z) \quad \text{and} \quad \text{key}_{\mathfrak{b}}^n(z) = \text{key}_{\mathfrak{a}}^n(z).$$

Consequently reservoir reindexing, product normalization, and allowable pullback do not change the key. The selection criterion in the M_n -construction is therefore the least $\text{key}_{\mathfrak{a}}^n(\langle x, y, m \rangle)$, not the least local name for y . The translation also has the following syntactic conventions.

1. Requirements that an area assignment, reservoir permutation, or canonical reindexing belong to L become the corresponding requirements over M_n .
2. A stage which in the L -construction codes (x, z, m) now uses the factor

$$\mathbb{A}^n(Y_n(\langle x, z, m \rangle, g_\eta)),$$

or the corresponding forcing name. Thus all stage forcings are instances of the coding mechanism from Subsection 4.5.

3. The role of Shoenfield absoluteness for a Π_2^1 coding witness is played by Theorem 4.9, since Ψ_n is Π_{n+2}^1 . The upward persistence of a forced nonmembership statement, which is Σ_{n+3}^1 , is supplied by Corollary 4.10.
4. The exactness assertion for the coding predicate is supplied by Theorem 4.47.

For completeness, we record the translated recursive notion. If

$$\mathbb{P}_\beta = \mathbb{C}_n^{\text{area}} * \dot{\mathbb{R}}_\beta$$

is an initial segment of an M_n -coding iteration and $\vec{\tau}$ is a finite tuple of stage-respecting names, an *admissible support pair* for $\vec{\tau}$ is a pair

$$(E, A) \in [U_\beta]^{\leq \omega} \times [\beta]^{\leq \omega}$$

such that E consists of reservoir coordinates, A consists of iteration stages, A is closed under the names defining the retained iterands, and the induced forcing on (E, A) is a complete subforcing of $\mathbb{P}_\beta$ over which every member of $\vec{\tau}$ is a name. We use the $\prec_n$ -least admissible pair. Likewise,

$$\mathbb{Q} \succeq_{\text{res}}^n \mathbb{P}$$

means that, after an M_n -coded reindexing of the reservoir onto a block of coordinates and an increasing reindexing of a closed set of stages, $\mathbb{P}$ is isomorphic to a complete subiteration of $\mathbb{Q}$. This is the $\mathfrak{t}_n$ -translation of the complete reservoir extension relation from Subsection 2.6.

Definition 4.48. A 0-allowable iteration over W_n is an M_n -coding iteration over the reservoir in the sense of Definition 4.44, paired with the canonical name for the empty auxiliary set.

Assume that $0 < \rho$ and that θ -allowability has been defined for every $\theta < \rho$. Let

$$F = \langle (\dot{x}_\beta, \dot{z}_\beta, \dot{m}_\beta) \mid \beta < \delta \rangle \in W_n$$

be stage-respecting bookkeeping, and let

$$\vec{\eta} = \langle \eta_\beta \mid \beta < \delta \rangle \in M_n$$

be injective, as in Definition 4.44. Put

$$U_\beta := \text{ran}(\vec{\eta} \restriction \beta).$$

At stage β , let (E_β, A_β) be the $\prec_n$ -least admissible support pair for the local bookkeeping names

$$(\dot{x}_\beta, \dot{z}_\beta, \dot{m}_\beta),$$

and require $E_\beta \subseteq U_\beta$. Put

$$\mathfrak{a}_\beta := (E_\beta, A_\beta).$$

Thus the new area coordinate η_β is not used by the local data. If $K * G_\beta$ is generic for the current initial segment, write

$$M_{\mathfrak{a}_\beta}^n = W_n[K_{E_\beta}][G_{A_\beta}]$$

and let x, z, m be the interpretations of the bookkeeping names in this local model.

For $\theta < \rho$ and $y \in M_{\mathfrak{a}_\beta}^n \cap 2^\omega$, say that y is θ -persistent for (x, m) over $\mathbb{P}_\beta$ if

$$(x, y) \in A_m^n$$

in the current extension and, for every M_n - θ -allowable iteration $\mathbb{Q} \in W_n$ together with a witness to

$$\mathbb{Q} \succeq_{\text{res}}^n \mathbb{P}_\beta,$$

the quotient over the selected copy of $\mathbb{P}_\beta$ forces

$$(x, y) \in A_m^n.$$

Let

$$C_\beta^{n,\rho} := \{(\theta, y) \in \rho \times (M_{\mathfrak{a}_\beta}^n \cap 2^\omega) : y \text{ is } \theta\text{-persistent for } (x, m) \text{ over } \mathbb{P}_\beta\}.$$

If $C_\beta^{n,\rho}$ is nonempty, choose first

$$\theta_\beta := \min\{\theta : \exists y ((\theta, y) \in C_\beta^{n,\rho})\},$$

and then choose y_β with $(\theta_\beta, y_\beta) \in C_\beta^{n,\rho}$ such that

$$\text{key}_{\mathfrak{a}_\beta}^n(\langle x, y_\beta, m \rangle)$$

is $\prec_n$ -least among all values of rank θ_β . Record $(x, y_\beta, m, \theta_\beta)$ in the auxiliary name. If $z \neq y_\beta$, put $z_\beta^* = z$. If $z = y_\beta$, let z_β^* be a real $w \neq y_\beta$ in $M_{\mathfrak{a}_\beta}^n$ for which

$$\text{key}_{\mathfrak{a}_\beta}^n(\langle x, w, m \rangle)$$

is $\prec_n$ -least. The next factor intentionally codes (x, z_β^*, m) , using $\prec_n$ -least stage-respecting names only to form the forcing name.

If $C_\beta^{n,\rho}$ is empty, the next factor intentionally codes the bookkeeping triple (x, z, m) and no tentative value is added. At limit stages take the finite-support limit of the almost disjoint tail and the union of the auxiliary names. All choices are made from $\prec_n$ and are uniform in the generic filter.

A forcing is M_n - ρ -allowable if it is isomorphic to the final forcing of an iteration obtained by this recursion. We denote the class of such forcings by

$$\mathcal{A}_\rho^{W_n}.$$

The definition is exactly the $\mathfrak{t}_n$ -translation of Definition 2.14. In particular, it uses the same stage-respecting support convention and the same complete-reservoir notion of continuation. A *relative M_n - ρ -allowable tail over $\mathbb{P}$* is a quotient $\mathbb{Q}/\dot{G}_\mathbb{P}$, together with its presentation, where $\mathbb{Q}$ is M_n - ρ -allowable and $\mathbb{Q} \succeq_{\text{res}}^n \mathbb{P}$.

Theorem 4.49. *The definitions and conclusions of Subsection 2.6 and Section 3 remain valid after applying $\mathfrak{t}_n$. More explicitly, the following statements hold in W_n .*

1. *If $\xi < \alpha$, then every M_n - α -allowable forcing is M_n - ξ -allowable. Hence*

$$\mathcal{A}_0^{W_n} \supseteq \mathcal{A}_1^{W_n} \supseteq \dots \supseteq \mathcal{A}_\alpha^{W_n} \supseteq \dots$$

2. *Relative allowable tails may be appended to an allowable iteration. If $\mathbb{P}^0$ and $\mathbb{P}^1$ are M_n - α -allowable, then*

$$\mathbb{P}^0 \times \mathbb{P}^1$$

is isomorphic to an M_n - α -allowable forcing.

3. *Tentative values remain in their intended sections. More precisely, suppose that $(\mathbb{P}, \dot{I})$ is M_n - α -allowable, G is $\mathbb{P}$ -generic over W_n , and*

$$(x, y, m, \xi) \in \dot{I}^G \quad (\xi < \alpha).$$

Then $W_n[G] \models (x, y) \in A_m^n$. Moreover, if $\mathbb{Q}$ is M_n - ξ -allowable and is supplied with a witness to $\mathbb{Q} \succeq_{\text{res}}^n \mathbb{P}$, then

$$W_n[G] \models “\mathbb{Q}/G \Vdash (x, y) \in A_m^n.”$$

The same conclusion holds for an ω_1 -length iteration whose proper initial segments are allowable.

4. Every class $\mathcal{A}_\alpha^{W_n}$ is nonempty, and the decreasing hierarchy stabilizes. If

$$\alpha_0^n := \min \left\{ \alpha : \forall \beta \geq \alpha \left(\mathcal{A}_\beta^{W_n} = \mathcal{A}_\alpha^{W_n} \right) \right\}$$

and

$$\alpha_n^* := \alpha_0^n + 1,$$

then the M_n - α_n^* -allowable forcings form the fixed class $\mathcal{A}_{\alpha_0^n}^{W_n}$. We call them M_n - ∞ -allowable. Consequently, if a local real $y \in A_{m,x}^n$ fails every persistence test of rank below α_n^* , there is an M_n - ∞ -allowable complete reservoir extension whose quotient forces $(x, y) \notin A_m^n$.

5. There is an ω_1 -length forcing

$$\mathbb{P}_{\omega_1}^n = \mathbb{C}_n^{\text{area}} * \dot{\mathbb{R}}_{\omega_1}^n$$

together with an auxiliary name $\dot{I}^n$ such that every countable initial segment is M_n - ∞ -allowable. The forcing has size $\aleph_1$, preserves cardinals and **CH**, and every real in its generic extension belongs to a countable intermediate stage. Whenever $K * G_{\omega_1}$ is generic over W_n , the final model

$$V_n^* := W_n[K][G_{\omega_1}]$$

satisfies

$$\Phi_n(u) \iff u \text{ was intentionally coded by } \mathbb{P}_{\omega_1}^n.$$

6. Writing

$$I^n := (\dot{I}^n)^{K * G_{\omega_1}},$$

exactly one of the following alternatives holds in V_n^* for every real x and every $m < \omega$:

- (a) if $(x, y, m, \xi) \in I^n$ for some y and $\xi < \alpha_n^*$, then there is a unique real y_0 such that

$$(x, y_0) \in A_m^n \quad \text{and} \quad \neg \Phi_n(\langle x, y_0, m \rangle);$$

every other member y of the section satisfies $\Phi_n(\langle x, y, m \rangle)$;

- (b) if no tentative value for (x, m) occurs in I^n , then

$$A_{m,x}^n = \emptyset.$$

Proof. We indicate precisely where the M_n -analogues established above supply the facts used in the L -construction.

First consider the forcing-theoretic part of the proof. The reservoir $\mathbb{C}_n^{\text{area}}$ has size $\aleph_1$, is ω -distributive over W_n , and admits every coordinate permutation belonging to M_n , by Lemma 4.31. Each coding factor is Knaster and has size $\aleph_1$, by Lemma 4.38. Admissible countable supports exist by the same nice-name and recursive-closure argument used over L : the reservoir adds no reals, the almost disjoint tail is c.c.c., and the countable supports are recursively enlarged so that every retained iterand is already named over the retained earlier stages and retained reservoir coordinates. Consequently all localized subiterations and quotient forcings used in the original proof are available over W_n .

The proofs of shrinking, the allowable-tail lemma, and nonemptiness are purely formal inductions on the rank and the length of the iteration. Replacing the bookkeeping instruction at a stage by the instruction actually used at a higher rank produces the lower-rank construction exactly as in the proof of Lemma 2.15.

For products, fix in M_n a partition

$$\omega_1 = B_0 \dot{\cup} B_1$$

and bijections $e_i : \omega_1 \rightarrow B_i$. Reindex the reservoirs of the two factors onto B_0 and B_1 and concatenate their finite-support tails. This normalizes the product over one reservoir. The induction from the proof of Lemma 2.16 applies verbatim: a real fails a persistence test in one factor precisely when it fails the corresponding test in the normalized product, because complete reservoir embeddings compose and because the induction hypothesis makes the product of the other factor with an eliminating forcing allowable. At stages where the persistent-value case applies, the translated local supports are related by an M_n -coded reservoir reindexing and an allowable pullback. Hence the local reals have the same persistence ranks, and the support-independence of key^n gives the same semantic keys in the original factor and in the normalized product. The product construction therefore makes the same stage choice. This proves (1) and (2).

The proof that tentative values remain in their sections is likewise the proof of Lemma 2.17. The selected value is persistent at its recorded rank, shrinking puts every relevant later initial segment into that rank, and complete reservoir extensions compose. For an ω_1 -length iteration whose proper initial segments are allowable, suppose toward a contradiction that membership fails. Write this Σ_{n+3}^1 statement as $\exists z \theta(z, x, y)$ with $\theta \in \Pi_{n+2}^1$. A name for the witness z has countable support and hence occurs in a bounded initial segment. Theorem 4.9 reflects $\theta(z, x, y)$ from the final extension to that bounded stage, contradicting persistence there. This proves (3).

Every allowable forcing has size at most $\aleph_1$ and belongs, up to isomorphism, to the set $H(\omega_2)^{W_n}$. Thus there are only set many isomorphism types. The

nonempty decreasing hierarchy must therefore stabilize, giving (4). The successor $\alpha_n^* = \alpha_0^n + 1$ is used so that the stage rule tests persistence against the fixed class itself.

We next transfer the final bookkeeping construction. Fix in W_n a map

$$F : \omega_1 \longrightarrow H(\omega_1)^{W_n}$$

which enumerates every stage-respecting system of names and every countable admissible enlargement of its support unboundedly often. Use the canonical area assignment $\eta_\beta = \beta$, which belongs to M_n . At a bookkeeping task for (x, z, m) , do nothing if the task is not legitimate or if the current extension does not satisfy $(x, z) \in A_m^n$. Otherwise form the local admissible support $\mathfrak{a}_\beta$ and apply the stage rule from Definition 4.48: if some real passes a persistence test below α_n^* , choose the least rank and then the real of $\prec_n$ -least key

$$\text{key}_{\mathfrak{a}_\beta}^n(\langle x, y, m \rangle).$$

If no real passes, failure at the fixed rank α_0^n supplies an M_n - ∞ -allowable complete reservoir extension whose quotient forces

$$(x, z) \notin A_m^n.$$

Restrict below the least condition forcing this statement and append the resulting relative tail. Such a tail may have countable length; concatenate the countable blocks and reindex them. Since a sum of ω_1 countable ordinals has order type at most ω_1 , the result has the displayed presentation $\mathbb{P}_{\omega_1}^n$. The induction just proved shows that every countable initial segment is M_n - ∞ -allowable. Lemma 4.45 gives the size and preservation assertions, and Theorem 4.47 gives the displayed equivalence with intentional coding. This proves (5).

It remains to check the final dichotomy. By the size and preservation statements already proved, every pair of successive intermediate models used below forms a small two-step extension in the sense of Definition 4.5. We use the following elementary consequence of Theorem 4.9. If $N \subseteq N[H]$ is one of the relevant two-step extensions, $x, y \in N$, and

$$N[H] \models (x, y) \in A_m^n,$$

then $N \models (x, y) \in A_m^n$. Indeed, write

$$\varphi_m^n(x, y) \equiv \forall r \, \vartheta(r, x, y)$$

with $\vartheta \in \Sigma_{n+2}^1$. For every $r \in N$, the extension satisfies $\vartheta(r, x, y)$, and Theorem 4.9 reflects this statement to N .

Suppose first that the auxiliary set contains a tentative value for (x, m) . Let $\xi_0 < \alpha_n^*$ be the least rank occurring for (x, m) . Among all occurrences

of rank ξ_0 , choose one whose semantic key is $\prec_n$ -least; write this key as κ_0 , let y_0 be the selected real, and let β be the first stage at which such an occurrence is added. Clause (3) gives $(x, y_0) \in A_m^n$ in the final model.

We use the following no-reversal observation. Suppose that, at a comparable stage with local support $\mathfrak{a}_\eta$, the construction considers the same pair (x, m) and selects $y' \neq y_0$ at rank $\xi' \leq \xi_0$. If $y_0 \in M_{\mathfrak{a}_\eta}$, then support-independence of key^n and stability of eliminability imply that y_0 is available there at rank ξ_0 . Hence the stage rule could choose y' only if either $\xi' < \xi_0$, or $\xi' = \xi_0$ and

$$\text{key}_{\mathfrak{a}_\eta}^n(\langle x, y', m \rangle) \prec_n \text{key}_{\mathfrak{a}_\eta}^n(\langle x, y_0, m \rangle).$$

By support-independence, the second key is κ_0 , and the first key is the key carried by the occurrence of y' at stage η . Thus both alternatives contradict the choice of (ξ_0, κ_0) among the occurrences in the final auxiliary set. No such reversal occurs.

It follows that the triple $\langle x, y_0, m \rangle$ is never intentionally coded. Before stage β , any coding of this triple would have resulted from selecting a different value for (x, m) with rank at most ξ_0 , which is excluded by the no-reversal observation. From stage β on, whenever y_0 is visible, the same observation forces the construction to protect y_0 rather than code it.

Let $y \neq y_0$ and suppose that $(x, y) \in A_m^n$ in the final model. At a sufficiently late bookkeeping stage, the local support contains x , y , and y_0 . The preceding downward-absoluteness observation shows that this local model already regards (x, y) as a member of A_m^n . The elimination case is impossible, because the resulting Σ_{n+3}^1 nonmembership would persist to the final model by Corollary 4.10. Hence the persistent-value case applies; by no reversal it protects y_0 , and therefore intentionally codes $\langle x, y, m \rangle$. Exactness of the coding predicate, supplied by Theorem 4.47, yields the first alternative in (6).

Finally suppose that no tentative value for (x, m) occurs and that, toward a contradiction, $(x, y) \in A_m^n$ in the final model. At a sufficiently late bookkeeping stage the names for x and y occur together. Downward absoluteness puts (x, y) in the section already at that stage. The persistent-value case would add a tentative value, so the elimination case must apply. It forces $(x, y) \notin A_m^n$, and this nonmembership persists to the final extension by Corollary 4.10, a contradiction. This proves (6) and completes the transfer. $\square$

For $m < \omega$, define

$$U_m^n(x, y) \iff (x, y) \in A_m^n \wedge \neg \Phi_n(\langle x, y, m \rangle).$$

Corollary 4.50. *In V_n^* , the relation U_m^n is a Π_{n+3}^1 uniformization of A_m^n for every $m < \omega$. Consequently, V_n^* satisfies the Π_{n+3}^1 -uniformization property.*

Proof. By Lemma 4.41, the predicate Φ_n is Σ_{n+3}^1 . Hence U_m^n is Π_{n+3}^1 , and Theorem 4.49(6) gives the domain and uniqueness requirements. The enumeration $\langle A_m^n \mid m < \omega \rangle$ may be chosen universal; real parameters are absorbed

into the first coordinate using the fixed recursive pairing function. Thus the conclusion holds for every boldface Π_{n+3}^1 relation. $\square$

Theorem 4.51. *Let $1 \leq n < \omega$, and assume that the canonical inner model M_n exists. Then there is a cardinal-preserving set-generic extension of M_n which satisfies CH and the Π_{n+3}^1 -uniformization property.*

Proof. First force over M_n with the branch product producing W_n , and then with $\mathbb{P}_{\omega_1}^n$. By Theorem 4.27 and Theorem 4.49(5), both steps preserve cardinals and CH. The resulting model is V_n^* , and the uniformization property follows from Corollary 4.50. $\square$

5 Further possible applications and open problems

We close with two directions suggested by the construction. First, the method should have an analogue in the generalized Baire-space context at a regular uncountable cardinal κ . In particular, one may ask whether the Π_1^1 -uniformization problem for $\kappa^{<\kappa}$ can consistently have a positive solution.

The methods in this paper leave the following question open:

Question 1. *Can the Π_n^1 -uniformization property be forced over L for $n > 3$?*

The method introduced here is so far limited to local effects. It would be interesting to force a less local, or even global, behaviour:

Question 2. *Given a pair $n, m \in \omega$ such that $n \neq m$ and $n \neq m + 1$, can one force a universe in which the Π_n^1 - and the Π_m^1 -uniformization properties both hold?*

Finally, it seems worthwhile to combine the present method with other forcing constructions in order to obtain models with properties that are usually consequences of determinacy assumptions. We record one paradigmatic question:

Question 3. *Assume some sufficiently big large cardinal notion which can exist in L . Is there a model which satisfies the Π_3^1 -uniformization property and in which every projective set of reals is Lebesgue measurable?*

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